47 JEE ADVANCED

YEARS (2024 TO 1978)

CHAPTER-WISE & TOPIC-WISE SOLVED PAPERS

FIRST TIME EVER!

Analyse the question-wise difficulty level in real-time with Correct (C), Wrong (W) and Unattempted (UA) questions response tagging provided by IIT-JEE

CHEMISTRY

ANSWER KEY VERIFIED FROM OFFICIAL WEBSITE OF JEE ADVANCED

JEE ADVANCED 2024 Paper Analysis

Paper 1

Section	Question Type	No. of Questions	Marks per Question	Negative marks per question	Total Marks (Section wise)
1	Single Correct MCQs	4	3	–1	12
2	More than One Correct MCQs	3	4	–2	12
3	Non Negative Integer Type Questions	6	4	0	24
4	Matrix Type Questions	4	3	–1	12
	Total Questions per subject	17			60

Paper 2

Section	Question Type	No. of Questions	Marks per Question	Negative marks per question	Total Marks (Section wise)
1	Single Correct MCQs	4	3	–1	12
2	More than One Correct MCQs	3	4	–2	12
3	Non Negative Integer Type Questions	6	4	0	24
4	(Numerical Value Based Questions)	4	3	–1	12
	Total Questions per subject	17			60

Total No. of Questions in Part 1 = 51 Questions
Total No. of Questions in Part 2 = 51 Questions
Each Subject Carries = 60 Marks
Total Marks = 180

Paper Analysis

Paper Difficulty Compared to Last Year	Easier Compared to 2023 Paper
Overall Difficulty Level of JEE 2024	
Easy	18%
Moderate	49%
Difficult	33%

Chemistry Topics	Paper 1 Difficulty Level	Paper 2 Difficulty Level	Comments/Remarks	
Physical	Paper 1 easier compared to last year	Paper2 easy to moderate compared to last year		No question from Ionic equilibrium question from every topic
Inorganic	Same	Slightly Tougher		No question from metallurgy
Organic	Overall Tougher	Tougher to moderate	No Questions from GOC, sterio-isomerism, nomenclature	1 Easy level question from combined chemistry in every day life and Biomolecule. Questions based on Name reaction + Polymer, Alkyl halide

Overall Analysis

Paper Difficulty Compared to Last Year
- Easy to moderate - Equal distribution among Physical, Organic, inorganic - in paper 1, slightly more Questions from organic - in paper 2, slightly more Questions from physical - 3 questions were difficult and out of the box - 1 question from paper1 from organic chemistry was out of the box - 2 question in paper 2 was out of the box - 5 questions from Paper 1 and 4 Question from paper 2 was very easy - Rest of the questions were moderate - Chemistry will decide the rank

JEE ADVANCED-5 Year (2023-19) Paper Analysis

Note: Due to unavailability of 2024 paper data we are unable to incorporate the 2024 analysis. As we are coming up with this book before the 2024 result.

Explanation of ★Unique★ Feature

C – Correct, W – Wrong, UA – Unattempted, PC – Partial Correct

C-31 W-22 UA-10 PC-36 represents the % of distribution of correct, wrong, unattempted and partial correct responses by students at any specific question in real time.(Data is taken from JEE Advanced website: https://jeeadv.ac.in/reports.html)

Classification helps students understand the varying **levels of difficulty**.

Easy	Medium	Hard
C > 20%	10% < C ≤ 20%	C ≤ 10%

For Example

22. Which of the following statement(s) is (are) correct regarding the root mean square speed (U_{rms}) and average translational kinetic energy (E_{av}) of a molecule in a gas at equilibrium?

C-31 W-22 UA-10 PC-36 (JEE Adv. 2019)

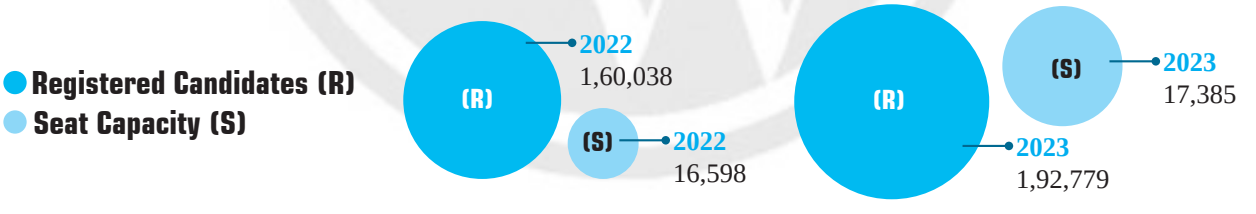
(*a*) U_{rms} is inversely proportional to the square root of its molecular mass

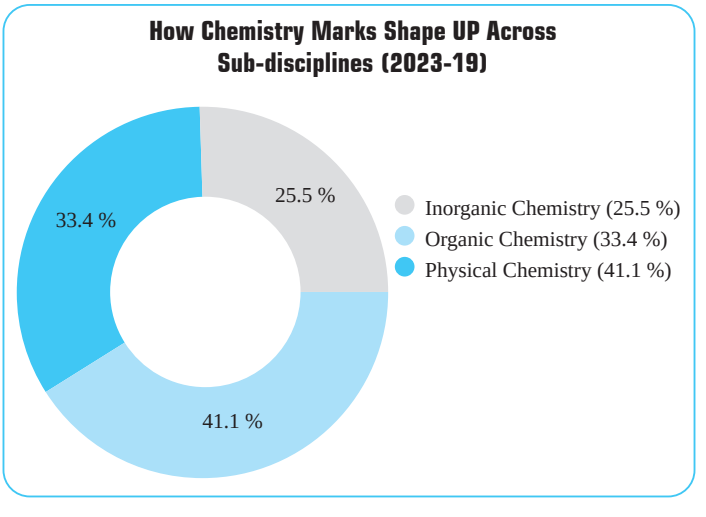
(*b*) U_{rms} is doubled when its temperature is increased four times

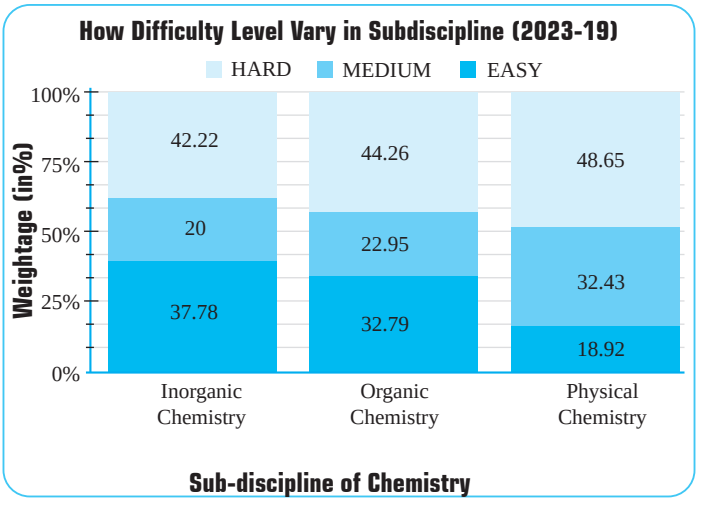
(*c*) E_{av} is doubled when its temperature is increased four times

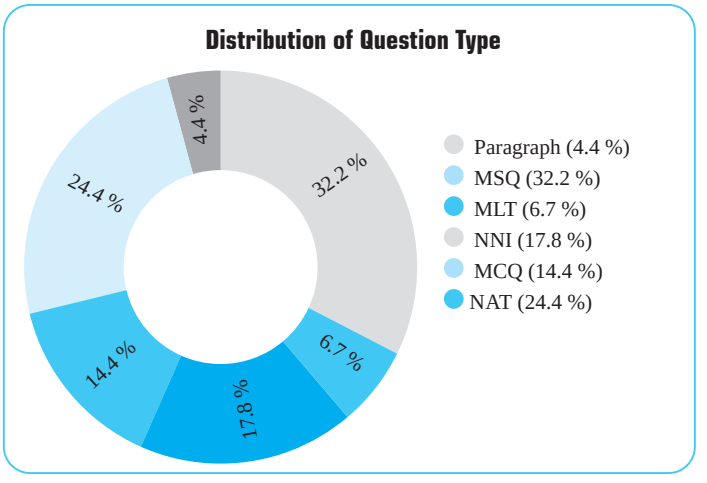
(*d*) E_{av} at a given temperature does not depend on its molecular mass

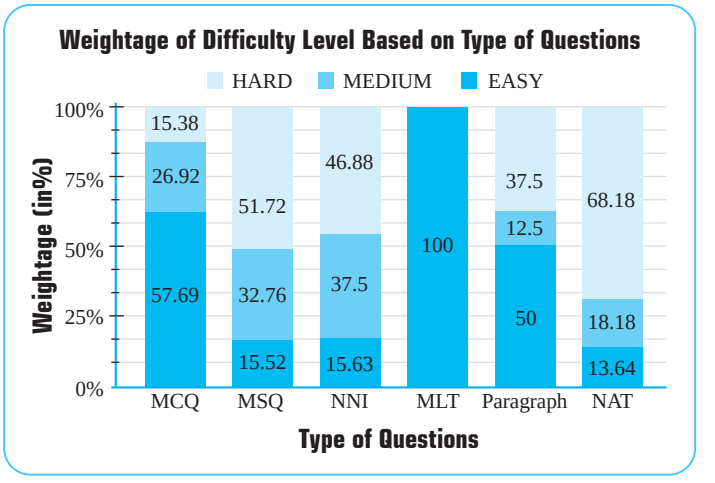
Question is considered EASY as Correct response recorded was greater than **20%**

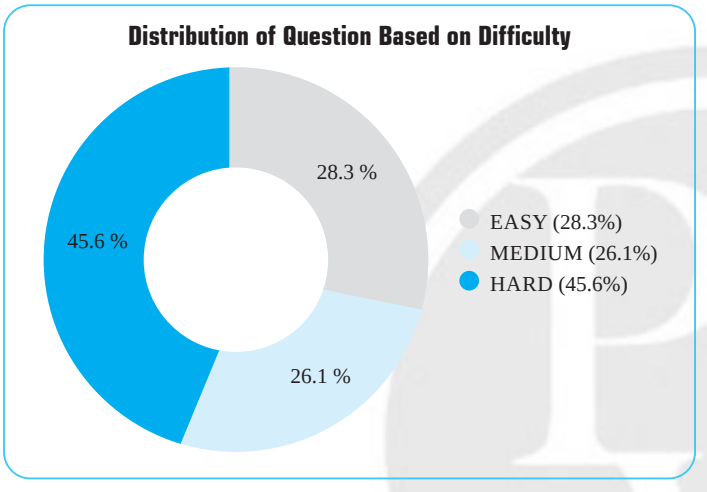


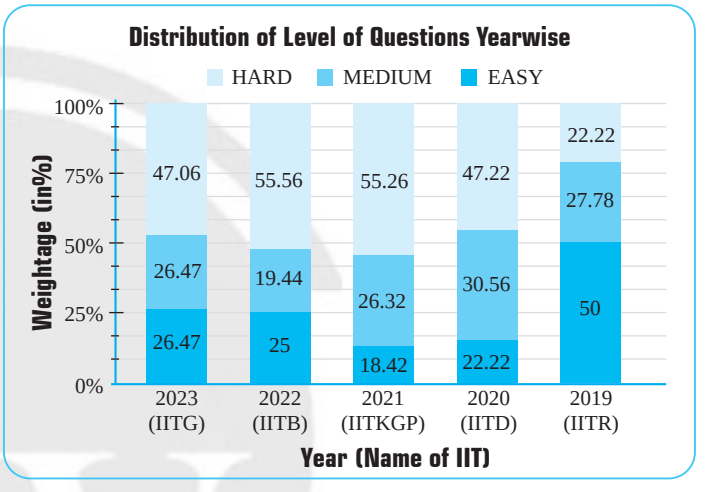








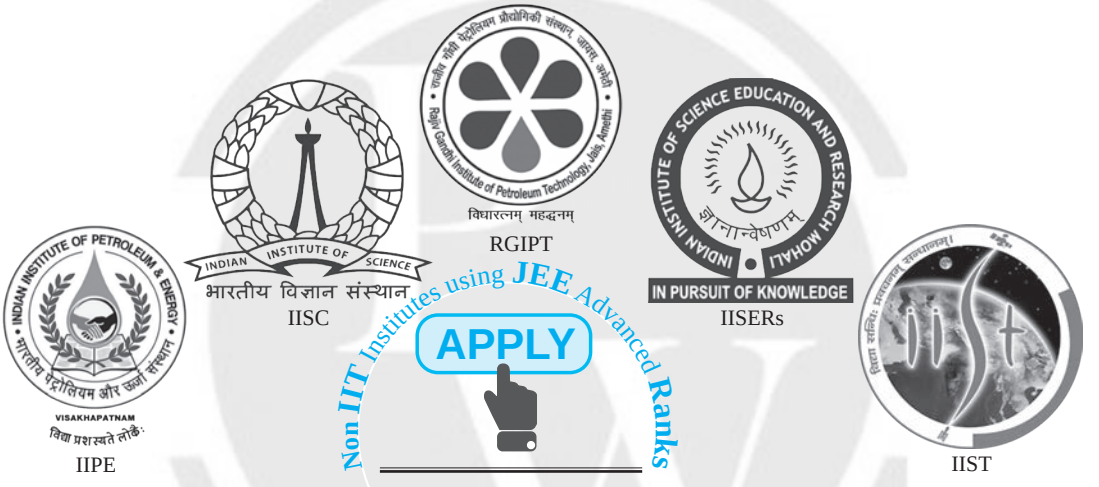


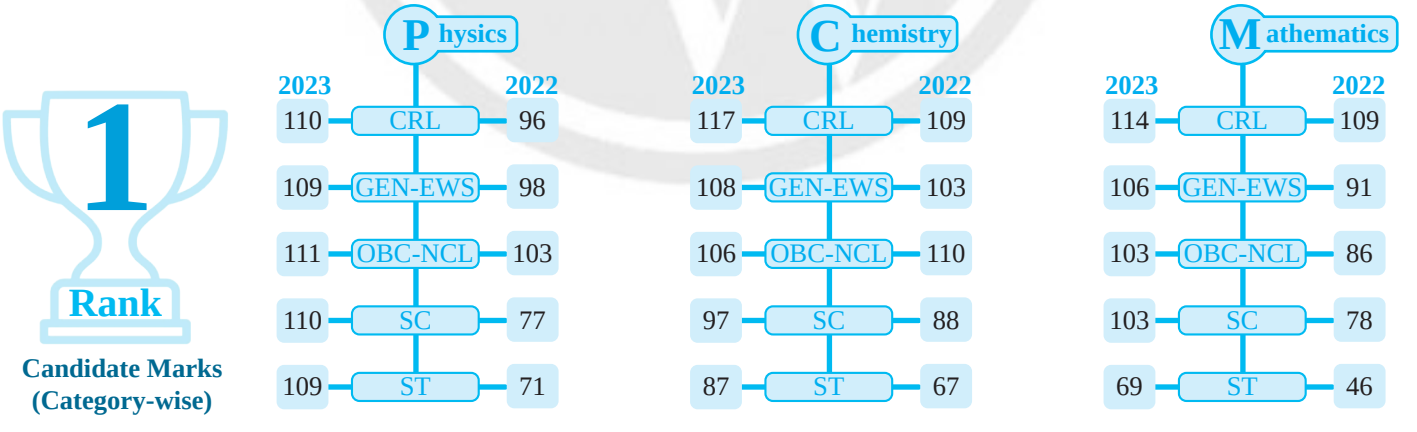


Chapter wise Weightage and count of Difficulty level of JEE Advanced Questions

Chapters Name	EASY	MEDIUM	HARD	Chapterwise Weightage(in %)
Aldehydes, Ketones and Carboxylic acids	6	3	6	**8.33**
Hydrocarbons	6	2	5	**7.22**
The d- and f-Block Elements	4	3	5	**6.67**
Thermodynamics	0	4	7	**6.11**
The p-Block Elements (Group 15 to 18)	4	2	4	**5.56**
Electrochemistry	0	3	6	**5**
Equilibrium	0	2	7	**5**
Coordination Compounds	4	2	3	**5**
Amines	2	2	5	**5**
General Principles and Processes of Isolation of Elements	1	2	5	4.44
Chemical Kinetics	3	1	4	4.44
Organic Chemistry-Some Basic Principles and Techniques	2	2	4	4.44
Chemical Bonding and Molecular Structure	3	2	1	3.33
Biomolecules	2	2	2	3.33
Structure of Atom	3	1	2	3.33
Solutions	2	1	3	3.33

Some Basic Concepts of Chemistry	0	3	2	2.78
The Solid State	0	5	0	2.78
Haloalkanes and Haloarenes	2	1	1	2.22
States of Matter	1	1	1	1.67
Surface Chemistry	2	0	1	1.67
Redox Reactions	0	1	2	1.67
Alcohols, Phenols and Ethers	0	0	3	1.67
The p-Block Elements (Group 13 and 14)	1	0	2	1.67
The s-Block Elements	2	0	0	1.11
Polymers	0	2	0	1.11
Chemistry in Everyday Life	0	0	1	0.56
Classification of Elements and Periodicity in Properties	1	0	0	0.56



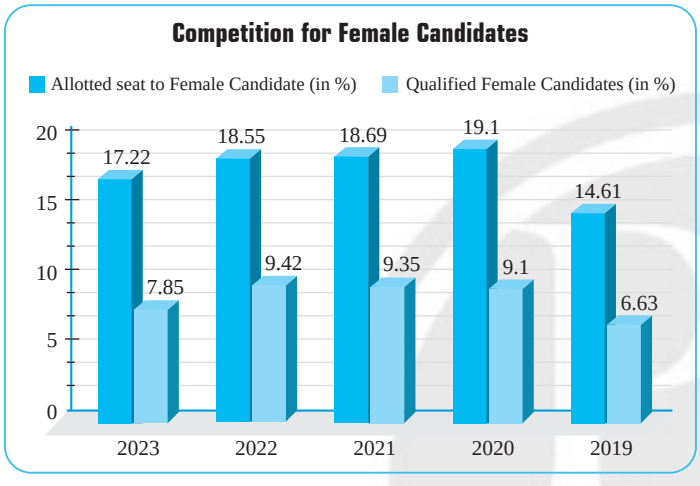


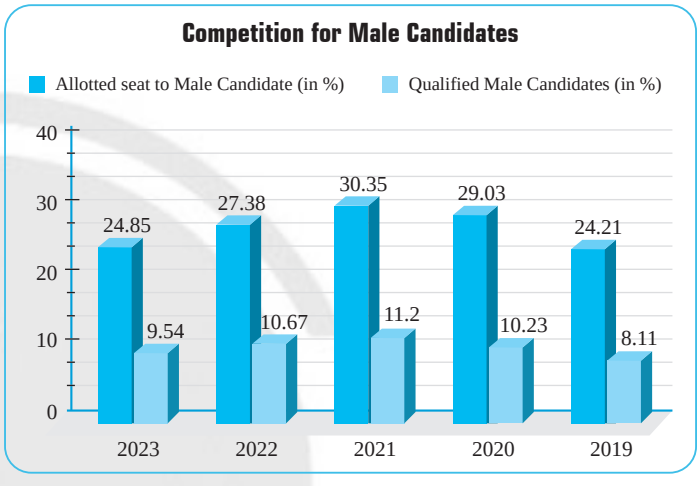
Climbing the IIT JEE Ladder: How Scores Translate to Ranks in General, OBC and SC Categories

Rank	Score (General)		Score (OBC)		Score (SC)	
	2023	2022	2023	2022	2023	2022
1	341	314	320	299	310	243
501	242	192	176	133	107	77
1001	219	170	152	114	87	60

1501	203	156	139	103	75	51
2001	191	146	130	95	68	46
2501	181	138	122	89	61	41
3001	174	132	116	83	56	37
3501	167	126	110	79	52	33
4001	162	122	106	75	49	30
4501	157	118	102	72	45	—
5001	153	115	98	69	43	—

Climbing the IIT Ladder: Tracking Candidate Success in JEE Advanced





2024 JEE ADVANCED SOLVED PAPER

Chemistry Paper-1

SECTION 1 (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (*a*), (*b*), (*c*) and (*d*). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –1 In all other cases.

1. A closed vessel contains 10 g of an ideal gas **X** at 300 K, which exerts 2 atm pressure. At the same temperature, 80 g of another ideal gas **Y** is added to it and the pressure becomes 6 atm. The ratio of root mean square velocities of **X** and **Y** at 300 K is

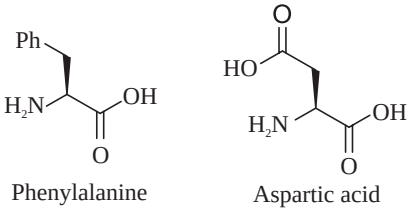
(*a*) $2\sqrt{2}:\sqrt{3}$ (*b*) $2\sqrt{2}:1$
(*c*) 1 : 2 (*d*) 2 : 1

2. At room temperature, disproportionation of an aqueous solution of *in situ* generated nitrous acid (HNO_2) gives the species

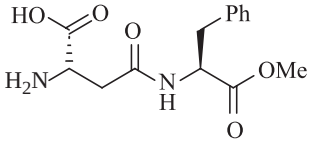
(*a*) H_3O^+, NO_3^- and NO (*b*) H_3O^+, NO_3^- and NO_2
(*c*) H_3O^+, NO^- and NO_2 (*d*) H_3O^+, NO_3^- and N_2O

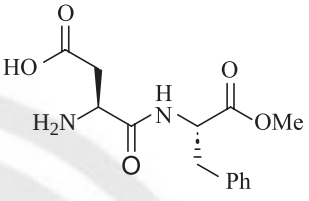
3. Aspartame, an artificial sweetener, is a dipeptide aspartyl phenylalanine methyl ester. The structure of aspartame is

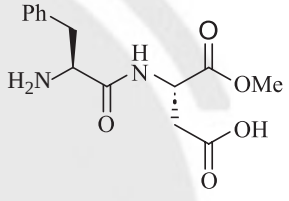
Structures of phenylalanine and aspartic acid are given below.

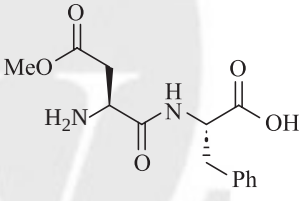


Phenylalanine Aspartic acid

(*a*) 

(*b*) 

(*c*) 

(*d*) 

4. Among the following options, select the option in which each complex in **Set-I** shows geometrical isomerism and the two complexes in **Set-II** are ionization isomers of each other.

[en = $H_2NCH_2CH_2NH_2$]

(*a*) **Set-I:** $[Ni(CO)_4]$ and $[PdCl_2(PPh_3)_2]$
Set-II: $[Co(NH_3)_5Cl]SO_4$ and $[Co(NH_3)_5(SO_4)]Cl$

(*b*) **Set-I:** $[Co(en)(NH_3)_2Cl_2]$ and $[PdCl_2(PPh_3)_2]$
Set-II: $[Co(NH_3)_6][Cr(CN)_6]$ and $[Cr(NH_3)_6][Co(CN)_6]$

(*c*) **Set-I:** $[Co(NH_3)_3(NO_2)_3]$ and $[Co(en)_2Cl_2]$
Set-II: $[Co(NH_3)_5Cl]SO_4$ and $[Co(NH_3)_5(SO_4)]Cl$

(*d*) **Set-I:** $[Cr(NH_3)_5Cl]Cl_2$ and $[Co(en)(NH_3)_2Cl_2]$
Set-II: $[Cr(H_2O)_6]Cl_3$ and $[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$

SECTION 2 (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options (*a*), (*b*), (*c*) and (*d*). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 **ONLY** if (all) the correct option(s) is(are) chosen;

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen;

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks: –2 In all other cases.

- For example, in a question, if (*a*), (*b*) and (*d*) are the **ONLY** three options corresponding to correct answers, then

choosing ONLY (*a*), (*b*) and (*d*) will get +4 marks;

choosing ONLY (*a*) and (*b*) will get +2 marks;

choosing ONLY (*a*) and (*d*) will get +2 marks;

choosing ONLY (*b*) and (*d*) will get +2 marks;

choosing ONLY (*a*) will get +1 mark;

choosing ONLY (*b*) will get +1 mark;

choosing ONLY (*d*) will get +1 mark;

choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get –2 marks.

5. Among the following, the correct statement(s) for electrons in an atom is(are)

(*a*) Uncertainty principle rules out the existence of definite paths for electrons.

(*b*) The energy of an electron in 2*s* orbital of an atom is lower than the energy of an electron that is infinitely far away from the nucleus.

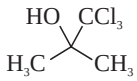
(*c*) According to Bohr's model, the most negative energy value for an electron is given by n = 1, which corresponds to the most stable orbit.

(*d*) According to Bohr's model, the magnitude of velocity of electrons increases with increase in values of n.

6. Reaction of *iso*-propylbenzene with O_2 followed by the treatment with H_3O^+ forms phenol and a by-product **P**. Reaction of **P** with 3 equivalents of Cl_2 gives compound **Q**. Treatment of **Q** with $Ca(OH)_2$ produces compound **R** and calcium salt **S**.

The correct statement(s) regarding **P**, **Q**, **R** and **S** is(are)

(*a*) Reaction of P with R in the presence of KOH followed by acidification gives



(*b*) Reaction of **R** with O_2 in the presence of light gives phosgene gas

(*c*) **Q** reacts with aqueous NaOH to produce Cl_3CCH_2OH and $Cl_3CCOONa$

(*d*) **S** on heating gives **P**

7. The option(s) in which at least three molecules follow Octet Rule is(are)

(*a*) CO_2, C_2H_4, NO and HCl

(*b*) NO_2, O_3, HCl and H_2SO_4

(*c*) BCl_3, NO, NO_2 and H_2SO_4

(*d*) CO_2, BCl_3, O_3 and C_2H_4

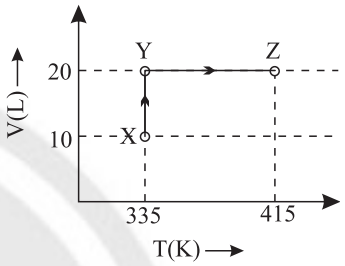
SECTION 3 (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If **ONLY** the correct integer is entered;

Zero Marks : 0 In all other cases.

8. Consider the following volume-temperature (V-T) diagram for the expansion of 5 moles of an ideal monoatomic gas.



Considering only P-V work is involved, the total change in enthalpy (in Joule) for the transformation of state in the sequence is **X → Y → Z** ____.

[Use the given data: Molar heat capacity of the gas for the given temperature range, $C_{V,m}$ = 12 J K^{-1} mol^{-1} and gas constant, R = 8.3 J K^{-1} mol^{-1}]

9. Consider the following reaction,

$$2H_2(g) + 2NO(g) \rightarrow N_2(g) + 2H_2O(g)$$

Which follows the mechanism given below:

$$2NO(g) \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} N_2O_2(g) \quad \text{(fast equilibrium)}$$

$$N_2O_2(g) + H_2(g) \xrightarrow{k_2} N_2O(g) + H_2O(g) \quad \text{(slow reaction)}$$

$$N_2O(g) + H_2(g) \xrightarrow{k_3} N_2(g) + H_2O(g) \quad \text{(fast reaction)}$$

The order of the reaction is_____?

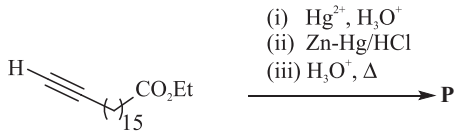
10. Complete reaction of acetaldehyde with excess formaldehyde, upon heating with conc. NaOH solution, gives **P** and **Q**. Compound **P** does not give Tollens' test, whereas **Q** on acidification gives positive Tollens' test. Treatment of **P** with excess cyclohexanone in the presence of catalytic amount of **P**-toluenesulfonic acid (PTSA) gives product **R**.

Sum of the number of methylene groups (– CH_2–) and oxygen atoms in **R** is_____.

11. Among $V(CO)_6$, $Cr(CO)_5$, $Cu(CO)_3$, $Mn(CO)_5$, $Fe(CO)_5$, $[Co(CO)_3]^{3-}$, $[Cr(CO)_4]^{4-}$, and $Ir(CO)_3$, the total number of species isoelectronic with $Ni(CO)_4$ is_____.

[Given, atomic number: V = 23, Cr = 24, Mn = 25, Fe = 26, Co = 27, Ni = 28, Cu = 29, Ir = 77]

12. In the following reaction sequence, the major product **P** is formed.



Glycerol reacts completely with excess **P** in the presence of an acid catalyst to form **Q**. Reaction of **Q** with excess NaOH followed by the treatment with $CaCl_2$ yields Ca-soap **R**, quantitatively. Starting with one mole of **Q**, the amount of **R** produced in gram is _____.

[Given, atomic weight: H = 1, C = 12, N = 14, O = 16, Na = 23, Cl = 35, Ca = 40]

13. Among the following complexes, the total number of diamagnetic species is_____.

$\left[Mn(NH_3)_6\right]^{3+}, \left[MnCl_6\right]^{3-}, \left[FeF_6\right]^{3-}, \left[CoF_6\right]^{3-}, \left[Fe(NH_3)_6\right]^{3+}$,

and $\left[Co(en)_3\right]^{3+}$

[Given, atomic number: Mn = 25, Fe = 26, Co = 27; en = $H_2NCH_2CH_2NH_2$]

SECTION 4 (Maximum Marks: 12)

- This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists: **List-I** and **List-II**.
- **List-I** has **Four** entries (P), (Q), (R) and (S) and **List-II** has **Five** entries (1), (2), (3), (4) and (5).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 **ONLY** if the option corresponding to the correct combination is chosen;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : –1 In all other cases.

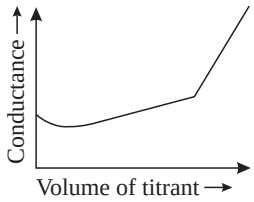
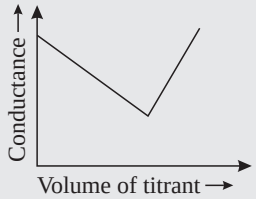
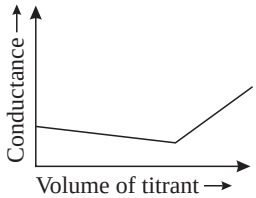
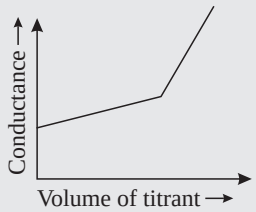
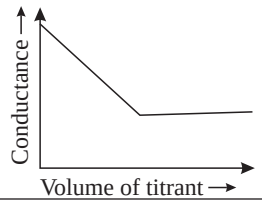
14. In a conductometric titration, small volume of titrant of higher concentration is added stepwise to a larger volume of titrate of much lower concentration, and the conductance is measured after each addition.

The limiting ionic conductivity (Λ_0) values (in mS m^2 mol^{-1}) for different ions in aqueous solutions are given below:

Ions	Ag^+	K^+	Na^+	H^+	NO_3^-	Cl^-	SO_4^{2-}	OH^-	CH_3COO^-
Λ_0	6.2	7.4	5.0	35.0	7.2	7.6	16.0	19.9	4.1

For different combinations of titrates and titrants given in **List-I**, the graphs of 'conductance' versus 'volume of titrant' are given in **List-II**.

Match each entry in **List-I** with the appropriate entry in **List-II** and choose the correct option.

List-I		List-II
(P) Titrate: KCl Titrant: $AgNO_3$	(1)	
(Q) Titrate: $AgNO_3$ Titrant: KCl	(2)	
(R) Titrate: NaOH Titrant: HCl	(3)	
(S) Titrate: NaOH Titrant: CH_3COOH	(4)	
	(5)	

(*a*) P-4, Q-3, R-2, S-5 (*b*) P-2, Q-4, R-3, S-1

(*c*) P-3, Q-4, R-2, S-5 (*d*) P-4, Q-3, R-2, S-1

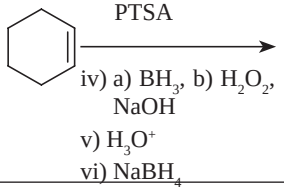
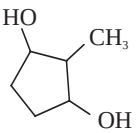
15. Based on **VSEPR** model, match the xenon compounds given in **List-I** with the corresponding geometries and the number of lone pairs on xenon given in **List-II** and choose the correct option.

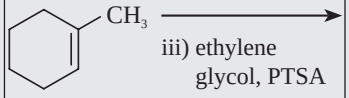
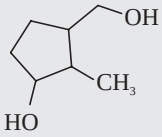
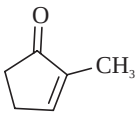
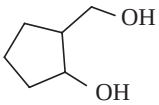
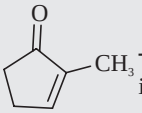
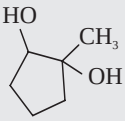
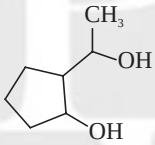
	List-I		List-II
(P)	XeF_2	(1)	Trigonal bipyramidal and two lone pair of electrons
(Q)	XeF_4	(2)	Tetrahedral and one lone pair of electrons
(R)	XeO_3	(3)	Octahedral and two lone pair of electrons
(S)	XeO_3F_2	(4)	Trigonal bipyramidal and no lone pair of electrons
		(5)	Trigonal bipyramidal and three lone pair of electrons

(*a*) P-5, Q-2, R-3, S-1 (*b*) P-5, Q-3, R-2, S-4

(*c*) P-4, Q-3, R-2, S-1 (*d*) P-4, Q-2, R-5, S-3

16. **List-I** contains various reaction sequences and **List-II** contains the possible products. Match each entry in **List-I** with the appropriate entry in **List-II** and choose the correct option.

	List-I		List-II
(P)		(1)	

(Q)	 i) O_3, Zn ii) aq. NaOH, Δ iii) ethylene glycol, PTSA iv) a) BH_3, b) H_2O_2, NaOH v) H_3O^+ vi) $NaBH_4$	(2)	
(R)	 i) ethylene glycol, PTSA ii) a) $Hg(OAc)_2$, H_2O, b) $NaBH_4$ iii) H_3O^+ iv) $NaBH_4$	(3)	
(S)	 i) ethylene glycol, PTSA ii) a) BH_3, b) H_2O_2, NaOH iii) H_3O^+ iv) $NaBH_4$	(4)	
		(5)	

(*a*) P-3, Q-5, R-4, S-1

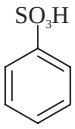
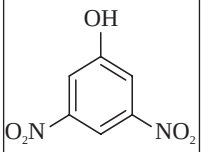
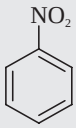
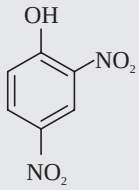
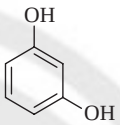
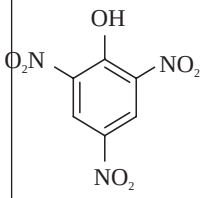
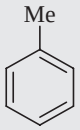
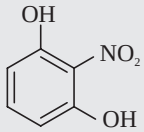
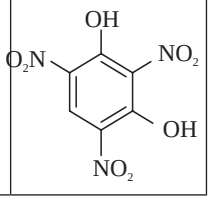
(*b*) P-3, Q-2, R-4, S-1

(*c*) P-3, Q-5, R-1, S-4

(*d*) P-5, Q-2, R-4, S-1

17. **List-I** contains various reaction sequences and **List-II** contains different phenolic compounds.

Match each entry in **List-I** with the appropriate entry in **List-II** and choose the correct option.

	List-I		List-II
(P)	 i) molten NaOH, H_3O^+ ii) Conc. HNO_3	(1)	
(Q)	 i) Conc. HNO_3 / Conc. H_2SO_4 ii) Sn / HCl iii) $NaNO_2$ /HCl, 0-5 °C, iv) H_2O v) Conc. HNO_3 / Conc. H_2SO_4	(2)	
(R)	 i) Conc. H_2SO_4 ii) Conc. HNO_3 iii) H_3O^+, Δ	(3)	
(S)	 i) a) $KMnO_4$ / KOH, Δ; b) H_3O^+ ii) Conc. HNO_3 / Conc. H_2SO_4, Δ iii) a) $SOCl_2$, b) NH_3 iv) Br_2, NaOH v) $NaNO_2$ / HCl, 0-5 °C vi) H_2O	(4)	
		(5)	

(*a*) P-2, Q-3, R-4, S-5 (*b*) P-2, Q-3, R-5, S-1

(*c*) P-3, Q-5, R-4, S-1 (*d*) P-3, Q-2, R-5, S-4

Solutions Paper-1

1. (*d*) Given,

$W_X = 10$ g

$P_X = 2$ atm

$W_Y = 80$ g

$P_Y = P_{total} - P_X$

$\Rightarrow 6 - 2 = 4$ atm

As $V_{rms} = \sqrt{\frac{3RT}{M}}$,

$$\frac{(V_{rms})_X}{(V_{rms})_Y} = \sqrt{\frac{\text{Molar mass of Y}}{\text{Molar mass of X}}} \quad ...(i)$$

As we know,

$PV = nRT$ (ideal gas equation)

Volume and temperature remains same.

$$P_X V = \frac{W_X}{M_X} RT$$

$$P_Y V = \frac{W_Y}{M_Y} RT$$

$$M_X \propto \frac{W_X}{P_X} \quad ....(ii)$$

$$M_Y \propto \frac{W_Y}{P_Y} \quad ....(iii)$$

put the value of M_X and M_Y from eq. (ii) and (iii) in eq. (i)

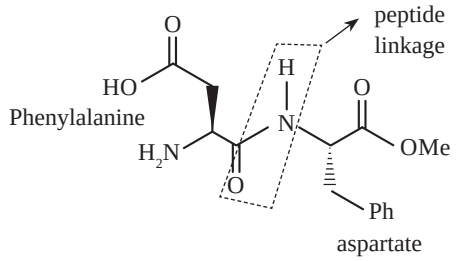
$$\frac{(V_{rms})_X}{(V_{rms})_Y} = \sqrt{\frac{W_Y}{P_Y} \cdot \frac{P_X}{W_X}}$$

$$= \sqrt{\frac{80}{4} \times \frac{2}{10}} = \sqrt{4} = \frac{2}{1} = 2:1$$

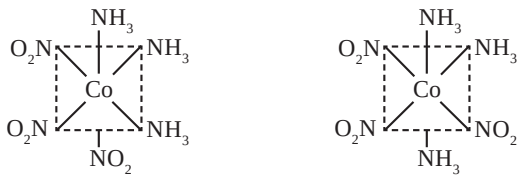
2. (*a*) $3\underset{+3}{HNO_2} \rightleftharpoons \underset{+5}{HNO_3} + \underset{+1}{2NO} + H_2O$ (Reduction: $HNO_2 \rightarrow 2NO$; oxidation: $HNO_2 \rightarrow HNO_3$)

H_3O^+, NO_3^- and NO

3. (*b*) Sol. Aspartame is



4. (*c*) Geometrical isomers of $[Co(NH_3)_3(NO_2)_3]$ are:



(Facial) (Meridional)

Geometrical isomers of $[Co(en)_2Cl_2]$ are:

(Cis) (Trans)

Set-II: $[Co(NH_3)_5Cl]SO_4$ and $[Co(NH_3)_5SO_4]Cl$ are ionisation isomers.

5. (*a,b,c*)

(*a*) The existence of definite paths or trajectories of electron and other similar particles is ruled out by Uncertainty principle.
So, option (*a*) is correct.

(*b*) A closer orbit or shell around the nucleus has less energy than a farther oribit
So, option (*b*) is also correct.

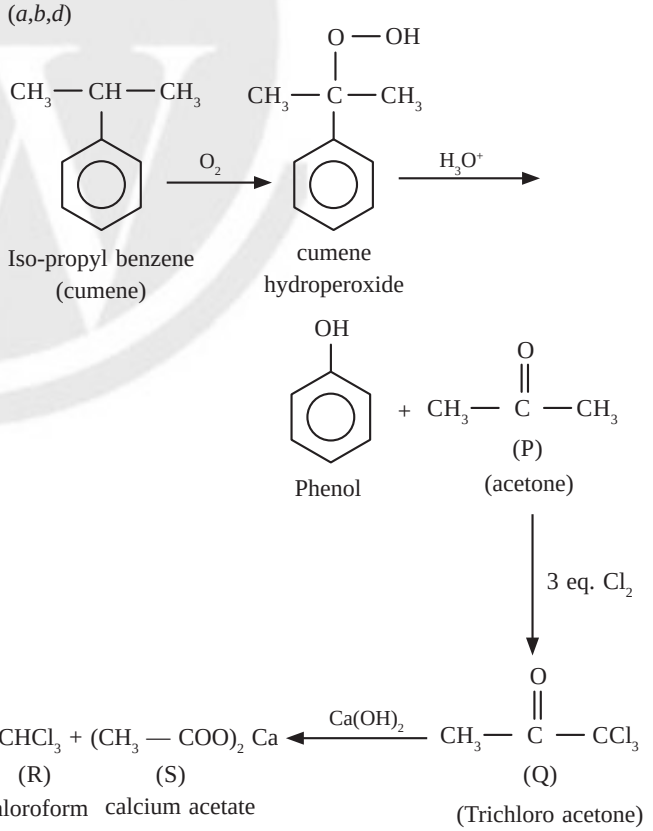
(*c*) $E = -13.6\frac{z^2}{n^2}$ eV / atom

So, $n = 1$ has most negative energy.
So, option (*c*) is also correct.

(*d*) $V = V_0 \times \frac{Z}{n}$

when n increases velocity decreases.
So, option (*d*) is incorrect.

6. (*a,b,d*)



(*a*) $CH_3-C(=O)-CH_3$ (P) + $CHCl_3$ (R) $\xrightarrow[\text{(ii) } H^+]{\text{(i) KOH}}$ $CH_3-C(OH)(CCl_3)-CH_3$

(*b*) $CHCl_3 + O_2 \xrightarrow{h\nu} COCl_2$
Phosgene

(*c*) $CH_3-\overset{\overset{O}{\|}}{C}-CCl_3 \xrightarrow{\text{aq. NaOH}} CH_3-\overset{\overset{O}{\|}}{C}-O^- + CHCl_3$

(*d*) $(CH_3-COO)_2Ca \xrightarrow{\Delta} CH_3-\overset{\overset{O}{\|}}{C}-CH_3$

7. (*a*,*d*)

Octet Rule: Tendency of atom to prefer eight electrons in their valence shell.

(*a*) CO_2, C_2H_4 and HCl follow octet rule.

(*b*) O_3 and HCl follow octet rule.

(*c*) None of them follow octet rule.

(*d*) CO_2, O_3 and C_2H_4 follow octet rule.

Correct answer is (*a*) and (*d*)

8. [8120]

$X \rightarrow Y$ is an isothermal process an ideal gas, as the temp is constant from X to Y.

$\Delta H = 0$

$Y \rightarrow Z$ is an isochoric process, as volume is constant from Y to Z.

$\therefore w = 0$

$\Delta U = nC_{v,m}(T_2 - T_1)$

$= 5 \times 12 (415 - 335)$

$= 4800$ J

$\Delta H = \Delta U + \Delta(PV)$ ($\because PV = nRT$ from ideal gas equation)

$= \Delta U + nR\Delta T$

$= 4800 + 5 \times 8.3 \times (415 - 335)$

$= 4800 + 3320$

$= 8120$ J

9. [3] Rate of reaction

$\Rightarrow r = k_2[N_2O_2][H_2]$ ...(i)

$\because$ Slowest step is the rate determining step

Now for intermediate $[N_2O_2]$,

$$\frac{k_1}{k_{-1}} = \frac{[N_2O_2]}{[NO]^2}$$

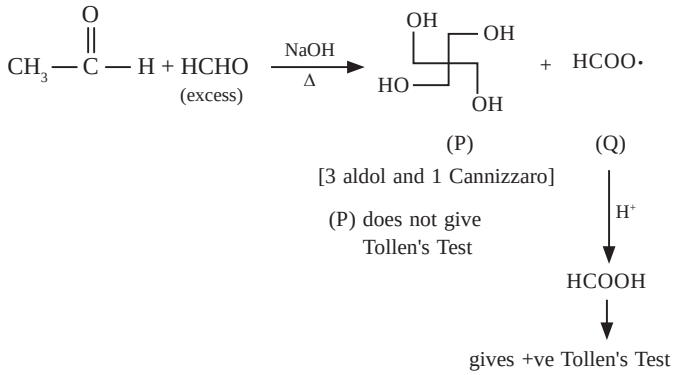
$$\Rightarrow [N_2O_2] = \frac{k_1}{k_{-1}}[NO]^2 \quad ...(ii)$$

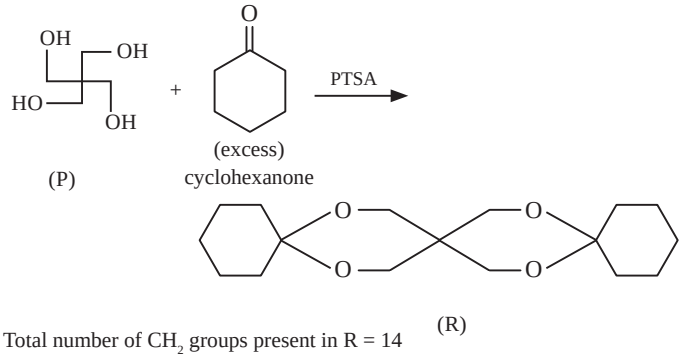
put the value of $[N_2O_2]$ from eq. (ii) to eq. (i)

$$r = \frac{k_2k_1}{k_{-1}}[NO]^2[H_2]$$

Overall order of reaction = 2 + 1 = 3

10. [18]





Total number of CH_2 groups present in R = 14

Total number of O-atoms present in R = 4

Required Answer = 14 + 4 = 18

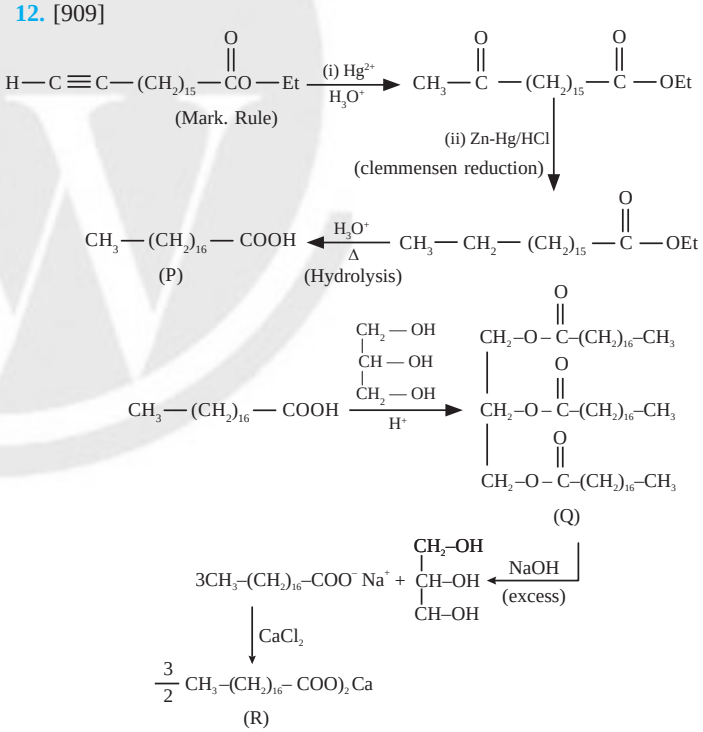
11. [1] Isoelectronic species are those having same no. of electrons.

Total number of electrons in

$Ni(CO)_4 = 28 + 4 \times 14 = 84$

Species	Total electron
$V(CO)_6 \rightarrow 23 + 6 \times 14$	= 107
$Cr(CO)_5 \rightarrow 24 + 5 \times 14$	= 94
$Cu(CO)_3 \rightarrow 29 + 3 \times 14$	= 71
$Mn(CO)_5 \rightarrow 25 + 5 \times 14$	= 95
$Fe(CO)_5 \rightarrow 26 + 5 \times 14$	= 96
$[Co(CO)_3]^{3-} \rightarrow 27 + 3 \times 14 + 3$	= 72
$[Cr(CO)_4]^{4-} \rightarrow 24 + 4 \times 14 + 4$	= 84
$Ir(CO)_3 \rightarrow 77 + 3 \times 14$	= 119

12. [909]



1 mole of Q produced 1.5 mole of R.

So, mass of R produced = 606 g × 1.5

= 909 g

13. [1]

	Electronic configuratin	Magnetic properties
$[Mn(NH_3)_6]^{3+}$	$[Ar]3d^4$	Paramagnetic
$[MnCl_6]^{3-}$	$[Ar]3d^4$	Paramagnetic
$[FeF_6]^{3-}$	$[Ar]3d^5$	Paramagnetic

$[CoF_6]^{3-}$	$[Ar]3d^6$	Paramagnetic
$[Fe(NH_3)_6]^{3+}$	$[Ar]3d^5$	Paramagnetic
$[Co(en)_3]^{3+}$	$[Ar]3d^6$	Diamagnetic

Only 1 complex is diamagnetic.

14. (*c*) (P) $KCl + AgNO_3 \rightarrow AgCl\downarrow + KNO_3$

Cl^- is replaced by NO_3^-

Conductance will slightly decrease till equivalence point, then it will increase

$P \rightarrow 3$

(Q) $AgNO_3 + KCl \rightarrow AgCl + KNO_3$

Ag^+ is replaced by K^+

Conductance will first increase upto equivalence point and then will increase

$Q \rightarrow 4$

(R) $NaOH + HCl \rightarrow NaCl + H_2O$

OH^- is replaced by Cl^-

$R \rightarrow 2$

(S) $NaOH + CH_3COOH$

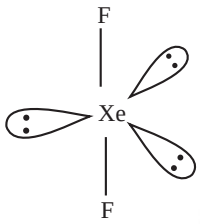
$\rightarrow CH_3CCONa + H_2O$

OH^- is replaced by CH_3COO^- conductance will first decrease and then become almost constant due to formation of buffer.

$S \rightarrow 5$

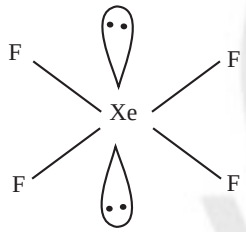
15. (*b*)

XeF_2:



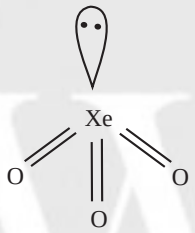
Hybridisation –sp^3d
Geometry: Trigonal bipyramidal
Number of lone pair = 3

XeF_4:



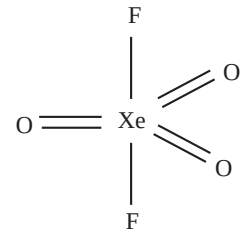
Hybridisation –sp^3d^2
Geometry: Octahedral
Number of lone pair = 2

XeO_3:



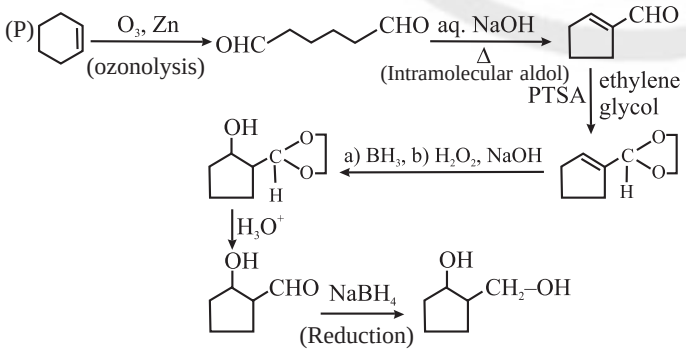
Hybridisation –sp^3
Geometry: tetrahedral
Number of lone pair = 1

XeO_3F_2:



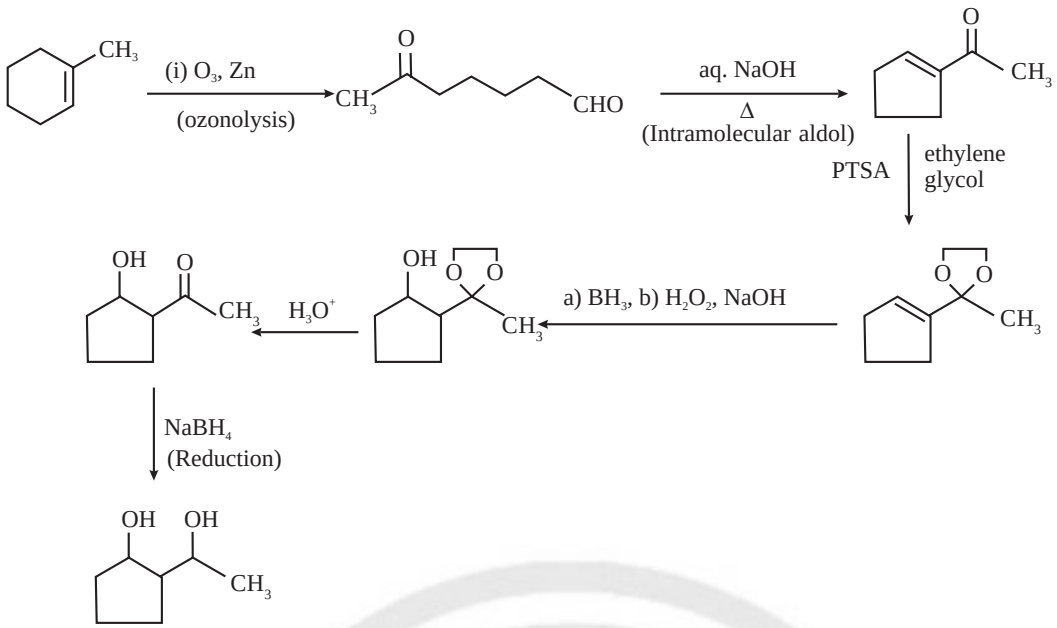
Hybridisation –sp^3d
Geometry: Trigonal bipyramidal
Number of lone pair = Zero

16. (*a*)



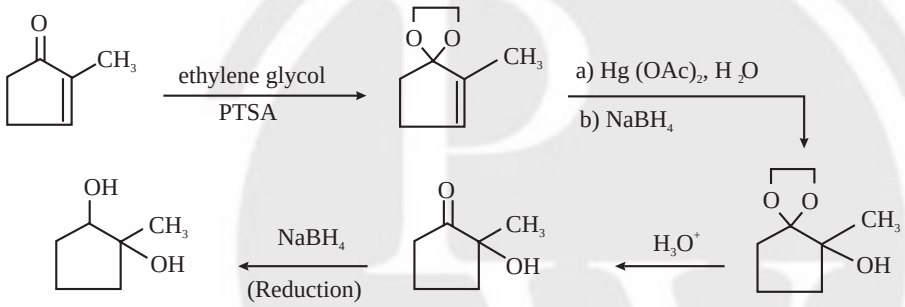
P-3

(Q)



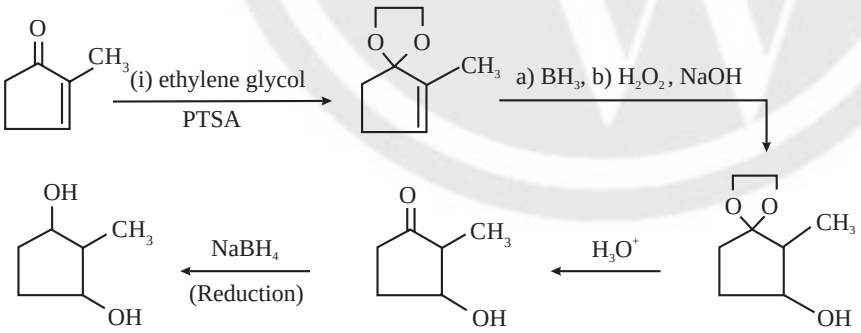
Q-5

(R)



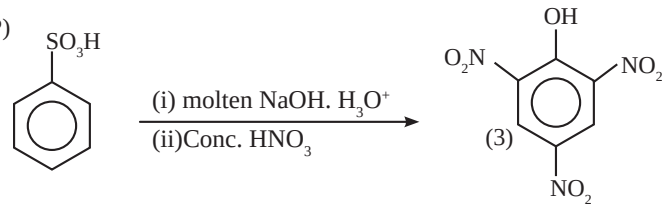
R-4

(S)

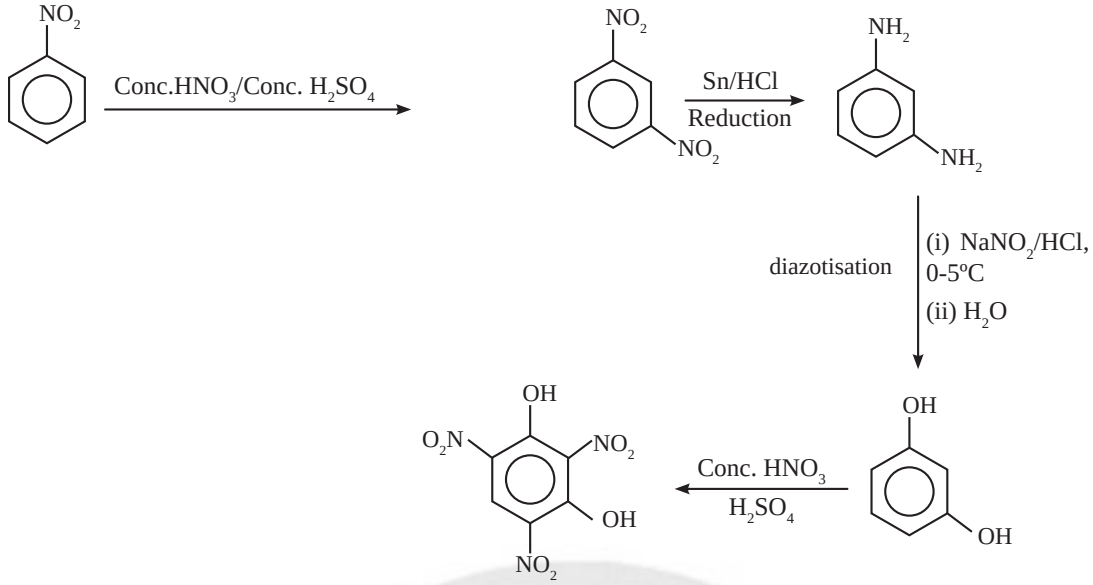


S-1

17. (*c*) (P)

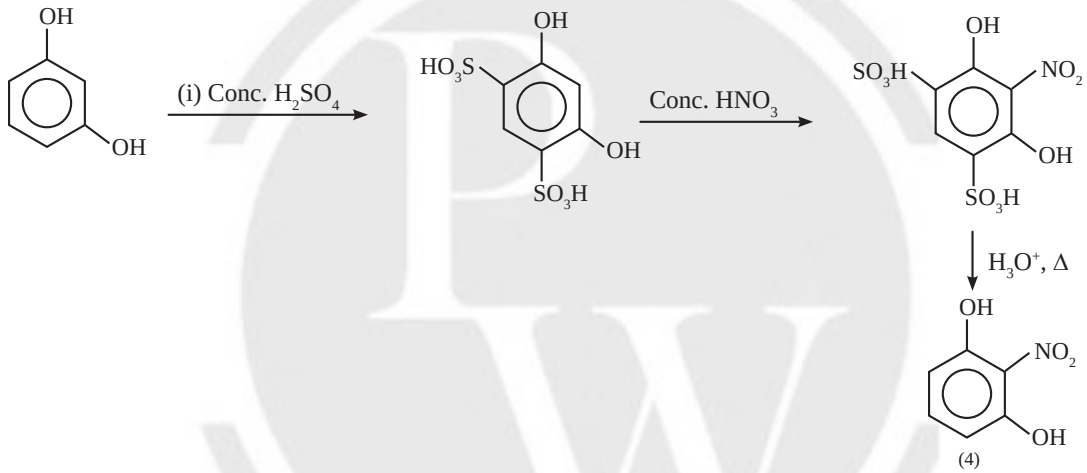


(Q)


NO2
Conc.HNO3/Conc. H2SO4
NO2
NO2
Sn/HCl
Reduction
NH2
NH2
diazotisation
(i) NaNO2/HCl,
0-5ºC
(ii) H2O
OH
OH
Conc. HNO3
H2SO4
OH
O2N
NO2
OH
NO2

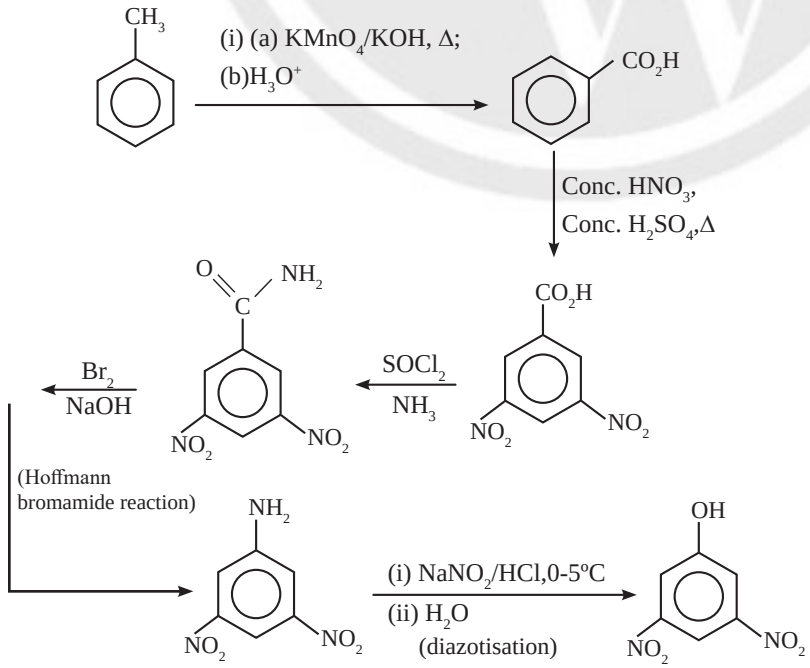
(5)

(R)


OH
OH
(i) Conc. H2SO4
OH
HO3S
OH
SO3H
Conc. HNO3
OH
SO3H
NO2
OH
SO3H
H3O+, Δ
OH
NO2
OH

(4)

(S)


CH3
(i) (a) KMnO4/KOH, Δ;
(b)H3O+
CO2H
Conc. HNO3,
Conc. H2SO4,Δ
CO2H
NO2
NO2
SOCl2
NH3
O
C
NH2
NO2
NO2
Br2
NaOH
(Hoffmann
bromamide reaction)
NH2
NO2
NO2
(i) NaNO2/HCl,0-5ºC
(ii) H2O
(diazotisation)
OH
NO2
NO2

1 SOME BASIC CONCEPTS OF CHEMISTRY

JEE-Advanced

Mole Concept, Percentage Composition, Molar Masses, and Empirical Formula

Single Correct

1. An aqueous solution of 6.3 g oxalic acid dihydrate is made up to 250 mL. The volume of 0.1 N NaOH required to completely neutralise 10 mL of this solution is **(IIT JEE 2001)**

   (*a*) 40 mL (*b*) 20 mL (*c*) 10 mL (*d*) 4 mL

2. The normality of 0.3M phosphorus acid (H_3PO_3) is **(IIT JEE 1999)**

   (*a*) 0.1 (*b*) 0.9 (*c*) 0.3 (*d*) 0.6

3. In which mode of expression, the concentration of a solution remains independent of temperature? **(IIT JEE 1998)**

   (*a*) Molarity (*b*) Normality
   (*c*) Formality (*d*) Molality

4. The volume strength of 1.5 NH_2O_2 is **(IIT JEE 1990)**

   (*a*) 4.8 (*b*) 8.4 (*c*) 3.0 (*d*) 8.0

5. A molal solution is one that contains one mole of solute in **(IIT JEE 1986)**

   (*a*) 1000 g of solvent (*b*) 1.0 L of solvent
   (*c*) 1.0 L of solution (*d*) 22.4 L of solution

6. If 0.50 mole of $BaCl_2$ is mixed with 0.20 mole of Na_3PO_4, the maximum number of moles of $Ba_3(PO_4)_2$ that can be formed is **(IIT JEE 1981)**

   (*a*) 0.70 (*b*) 0.50 (*c*) 0.20 (*d*) 0.10

7. The total number of electrons in one molecule of carbon dioxide is **(IIT JEE 1979)**

   (*a*) 22 (*b*) 44 (*c*) 68 (*d*) 88

Numerical/Integer Type

8. Aluminium reacts with sulphuric acid to form aluminium sulphate and hydrogen. What is the volume of hydrogen gas in litre (L) produced at 300 K and 1.0 atm pressure, when 5.4 g of aluminium and 50.0 mL of 5.0 M sulphuric acid are combined for the reaction? (Use molar mass of aluminium as 27.0 g mol^{-1}, R = 0.082 atm L mol^{-1} K^{-1}) **C-18.21 W-64.19 UA-17.6 PC-0 (JEE Adv. 2020)**

9. The mole fraction of urea in an aqueous urea solution containing 900 g of water is 0.05. If the density of the solution is 1.2 g cm^{-3}, the molarity of urea solution is ___.
   (Given data: Molar masses of urea and water are 60 g mol^{-1} and 18 g mol, respectively)
   C-17.05 W-70.35 UA-12.61 (JEE Adv. 2019)

10. The mole fraction of a solute in a solution is 0.1. At 298 K, molarity of this solution is the same as its molality. Density of this solution at 298 K is 2.0 g cm^{-3}. The ratio of the molecular weights of the solute and solvent, $\left(\frac{m_{solute}}{m_{solvent}}\right)$ is ...
    C-28.66 W-66.23 UA-5.11 (JEE Adv. 2016)

11. A compound H_2X with molar weight of 80 g is dissolved in a solvent having density of 0.4 g mL^{-1}. Assuming no change in volume upon dissolution, the molality of a 3.2 molar solution is
    C-39.97 W-51.56 UA-8.47 (JEE Adv. 2014)

12. 29.2% (w/W) HCl stock solution has density of 1.25 g mL^{-1}. The molecular weight of HCl is 36.5 g mol^{-1}. The volume (mL) of stock solution required to prepare a 200 mL solution 0.4 M HCl is
    C-15.53 W-67.67 UA-16.8 (IIT JEE 2012)

Fill in the Blanks

13. 3.0 g of a salt of molecular weight 30 is dissolved in 250 g water. The molality of the solution is ..... **(IIT JEE 1983)**

14. The total number of electrons present in 18 mL of water is ..... **(IIT JEE 1980)**

Subjective

15. 20% surface sites have adsorbed N_2. On heating N_2 gas evolved from sites and were collected at 0.001 atm and 298 K in a container of volume is 2.46 cm^3. Density of surface sites is $6.023 \times 10^{14}/cm^2$ and surface area is 1000 cm^2, find out the number of surface sites occupied per molecule of N_2. **(IIT JEE 2005)**

16. Calculate the amount of calcium oxide required when it reacts with 852 g of P_4O_{10}. **(IIT JEE 2005)**

17. In a solution of 100 mL 0.5 M acetic acid, one gram of active charcoal is added, which adsorbs acetic acid. It is found that the concentration of acetic acid becomes 0.49M. If surface area of charcoal is 3.01×10^2 m^2, calculate the area occupied by a single acetic acid molecule on the surface of charcoal. **(IIT JEE 2003)**

18. Find the molarity of water. Given: $\rho = 1000$ kg/m^3 **(IIT JEE 2003)**

19. How many millilitres of 0.5M H_2SO_4 are needed to dissolve 0.5 g of copper (II) carbonate? **(IIT JEE 1999)**

20. An aqueous solution containing 0.10g KIO_3 (formula weight = 214.0) was treated with an excess of KI solution. The solution was acidified with HCl. The liberated I_2 consumed 45.0 mL of thiosulphate solution decolourise the blue starch-iodine complex. Calculate the molarity of the sodium thiosulphate solution. **(IIT JEE 1998)**

21. To a 25 mLH_2O_2 solution, excess of acidified solution of potassium iodide was added. The iodine liberated required 20 mL of 0.3 N sodium thiosulphate solution. Calculate the volume strength of H_2O_2 solution **(IIT JEE 1997)**

22. A 5.0 cm^3 solution of H_2O_2 liberates 0.508 g of iodine from an acidified KI solution. Calculate the strength of H_2O_2 solution in terms of volume strength at STP. **(IIT JEE 1995)**

23. 8.0575×10^{-2} kg of Glauber's salt is dissolved in water to obtain 1 dm^3 of solution of density 1077.2 kg m^{-3}. Calculate the molality, molarity and mole fraction of Na_2SO_4 in solution. **(IIT JEE 1994)**

24. Upon mixing 45.0 mL, 0.25 M lead nitrate solution with 25.0 mL of a 0.10 M chromic sulphate solution, precipitation of lead sulphate takes place. How many moles of lead sulphate are formed? Also calculate the molar concentrations of species left behind in the final solution. Assume that lead sulphate is completely insoluble. **(IIT JEE 1993)**

25. Calculate the molality of 1.0 L solution of 93% H_2SO_4, (weight/volume). The density of the solution is 1.84 g/mL. **(IIT JEE 1990)**

26. n-butane is produced by monobromination of ethane followed by Wurtz's reaction. Calculate volume of ethane at NTP required to produce 55 g n-butane, if the bromination takes place with 90% yield and the Wurtz's reaction with 85% yield. **(IIT JEE 1989)**

27. A sugar syrup of weight 214.2 g contains 34.2 g of sugar ($C_{12}H_{22}O_{11}$). Calculate (i) molal concentration and (ii) mole fraction of sugar in syrup **(IIT JEE 1988)**

28. An unknown compound of carbon, hydrogen and oxygen contains 69.77%C and 11.63%H and has a molecular weight of 86. It does not reduce Fehling's solution but forms a bisulfite addition compound and gives a positive iodoform test. What is the possible structure(s) of unknown compounds? **(IIT JEE 1987)**

29. The density of a 3M sodium thiosulphate solution ($Na_2S_2O_3$) is 1.25 g per mL. Calculate (i) the percentage by weight of sodium thiosulphate (ii) the mole fraction of sodium thiosulphate and (iii) the molalities of Na^+ and $S_2O_3^{2-}$ ions. **(IIT JEE 1983)**

30. In the analysis of 0.5 g sample of feldspar, a mixture of chlorides of sodium and potassium is obtained, which weighs 0.1180 g. Subsequent treatment of the mixed chlorides with silver nitrate gives 0.2451 g of silver chloride. What is the percentage of sodium oxide and potassium oxide in the sample? **(IIT JEE 1979)**

31. The vapor density (hydrogen = 1) of a mixture consisting of NO_2 and N_2O_4 is 38.3 at 26.7°C. Calculate the number of moles of NO_2 in 100 g of the mixture. **(IIT JEE 1979)**

Uncertainty in Measurement and Laws of Chemical Combinations, Atomic and Molecular Masses, Stoichiometry and Stoichiometric Calculation

Single Correct

32. 2.76 g of silver carbonate on being strongly heated yields a residue weighing **(IIT JEE 1979)**

(*a*) 2.16 g (*b*) 2.48 g (*c*) 2.32 g (*d*) 2.64 g

33. When the same amount of zinc is treated separately with excess of sulphuric acid and excess of sodium hydroxide, the ratio of volumes of hydrogen evolved is **(IIT JEE 1979)**

(*a*) 1:1 (*b*) 1:2

(*c*) 2:1 (*d*) 9 : 4

34. The largest number of molecules is in **(IIT JEE 1979)**

(*a*) 36 g of water

(*b*) 28 g of CO

(*c*) 46 g of ethyl alcohol

(*d*) 54 g of nitrogen pentoxide (N_2O_5)

35. A gaseous mixture contains oxygen and nitrogen in the ratio of 1:4 by weight. Therefore, the ratio of their number of molecules is **(IIT JEE 1979)**

(*a*) 1:4 (*b*) 1:8 (*c*) 7:32 (*d*) 88

Multiple Correct

36. To check the principle of multiple proportions, a series of pure binary compounds (P_mQ_n) were analyzed and their composition is tabulated below. The correct option(s) is(are)

C-14.77 W-26.73 UA-39.64 (JEE Adv. 2022)

Compound	Weight % of P	Weight % of Q
1	50	50
2	44.4	55.6
3	40	60

(*a*) If empirical formula of compound 3 is P_3Q_4. then the empirical formula of compound 2 is P_3Q_5.

(*b*) If empirical formula of compound 3 is P_3Q_2 and atomic weight of clement P is 20, then the atomic weight of Q is 45.

(*c*) If empirical formula of compound 2 is PQ, then the empirical formula of the compound 1 is P_5Q_4.

(*d*) If atomic weight of P and Q are 70 and 35, respectively, then the empirical formula of compound 1 is P_2Q.

Numerical/Integer Type

37. The stoichiometric reaction of 516 g of dimethyldi-chlorosilane with water results in a tetrameric cyclic product X in 75% yield. The weight (in g) of X obtained is___.
[Use, molar mass (g mol^{-1}): H = 1, C = 12, O = 16, Si = 28, Cl = 35.5]

C-3.16 W-59.71 UA-37.13 (JEE Adv. 2023)

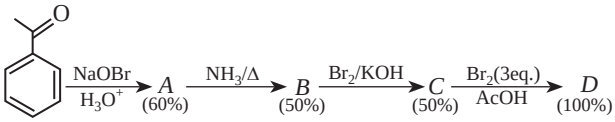
38. Dissolving 1.24 g of white phosphorous in boiling NaOH solution in an inert atmosphere gives gas Q. The amount of $CuSO_4$ (in g) required to completely consume the gas Q is____.
[Given: Atomic mass of H = 1, 0 = 16, Na = 23, P = 31, S = 32, Cu = 63] **C1.78 W53.98 UA44.24 (JEE Adv. 2022)**

39. Galena (an ore) is partially oxidized by passing air through it at high temperature. After some time, the passage of air is stopped, but the heating is continued in a closed furnace such that the contents undergo self-reduction. The weight (in Kg) of Pb produced per kg of O_2 consumed is
(Atomic weights in g/mol :O = 16, S = 32, Pb = 207)

C-8.91 W-71.9 UA-19.2 (JEE Adv. 2018)

40. In the following reaction sequence, the amount of D (in gram) formed from 10 moles of acetophenone is .....
(Atomic weights in g mol^{-1} : H = 1, C = 12, N = 14, O = 16, Br = 80. The yield (%) corresponding to the product in each step is given in the parenthesis)

C-7.27 W-72.1 UA-20.63 (JEE Adv. 2018)



41. A student performs a titration with different burettes and finds titrate values of 25.2 mL, 25.25 mL, and 25.0 mL. The number of significant figures in the average titrate value is **(IIT JEE 2010)**

Fill in the Blanks

42. The compound $YBa_2Cu_3O_7$, which shows super conductivity, has copper in oxidation state ..... Assume that the rare earth element yttrium is in its usual +3 oxidation state. **(IIT JEE 1994)**

43. The weight of 1×10^{22} molecules of $CuSO_4{\cdot}5H_2O$ is _____ **(IIT JEE 1991)**

44. The modern atomic mass unit is based on the mass of _____ **(IIT JEE 1980)**

Subjective

45. A plant virus is found to consist of uniform cylindrical particles of 150 in diameter and 5000 angstrom long. The specific volume of the virus is 0.75 cm^3/g. If the virus is considered to be a single particle, find its molar mass. **(IIT JEE 1999)**

46. A 3.00 g sample containing Fe_3O_4, Fe_2O_3 and an inert impure substance, is treated with excess of KI solution in presence of dilute H_2SO_4. The entire iron is converted into Fe^{2+} along with the liberation of iodine. The resulting solution is diluted to 100 mL. A 20 mL of the diluted solution requires 11.0 mL of 0.5M$Na_2S_2O_3$ solution to reduce the iodine present. A 50 mL of the dilute solution, after complete extraction of the iodine required 12.80 mL of 0.25M$KMnO_4$ solution in dilute H_2SO_4 medium for the oxidation of Fe^{2+}. Calculate the percentage of Fe_2O_3 and Fe_3O_4 in the original sample. **(IIT JEE 1996)**

47. A 20.0 cm^3 mixture of CO, CH_4 and He gases is exploded by an electric discharge at room temperature with excess of oxygen. The volume contraction is found to be 13.0 cm^3. A further contraction of 14.0 cm^3 occurs when the residual gas is treated with KOH solution. Find out the composition of the gaseous mixture in terms of volume percentage. **(IIT JEE 1995)**

48. A is a binary compound of a univalent metal. 1.422 g of A reacts completely with 0.321 g of sulphur in an evacuated and sealed tube to give 1.743 g of a white crystalline solid B, that forms a hydrated double salt, C with $Al_2(SO_4)_3$. Identify A, B and C. **(IIT JEE 1994)**

49. One gram of commercial $AgNO_3$ is dissolved in 50 mL of water. It is treated with 50 mL of a KI solution. The silver iodide thus precipitated is filtered off. Excess of KI in the filtrate is titrated with (M/10) KIO_3 solution in presence of 6 M HCl till all I^- ions are converted into ICl. It requires 50 mL of (M/10) KIO_3 solution, 20 mL of the same stock solution of KI requires 30 mL of (M/10) KIO_3 under similar conditions. Calculate the percentage of $AgNO_3$ in the sample.
Reaction $KIO_3 + 2KI + 6HCl \rightarrow 3ICl + 3KCl + 3H_2O$ **(IIT JEE 1991)**

50. A solid mixture (5.0 g) consisting of lead nitrate and sodium nitrate was heated below 600°C until the weight of the residue was constant. If the loss in weight is 28.0 per cent, find the amount of lead nitrate and sodium nitrate in the mixture. **(IIT JEE 1990)**

51. 1.0 L of a mixture of CO and CO_2 is taken. This mixture is passed through a tube containing red hot charcoal. The volume now becomes 1.6 L. The volumes are measured under the same conditions. Find the composition of the mixture by volume. **(IIT JEE 1980)**

52. 5.00 mL of a gas containing only carbon and hydrogen were mixed with an excess of oxygen (30 mL) and the mixture exploded by means of electric spark. After explosion, the volume of the mixed gases remaining was 25 mL. On adding a concentrated solution of KOH, the volume further diminished to 15 mL, the residual gas being pure oxygen. All volumes have been reduced to NTP. Calculate the molecular formula of the hydrocarbon gas. **(IIT JEE 1979)**

53. Accounts for the following. Limit your answer to two sentences, "Atomic weights of most of the elements are fractional". **(IIT JEE 1979)**

ANSWER KEY

1. (*a*)	**2.** (*d*)	**3.** (*d*)	**4.** (*b*)	**5.** (*a*)	**6.** (*d*)	**7.** (*a*)	**8.** [6.15]	**9.** [2.98 or 2.99]
10. [9]	**11.** [8]	**12.** [8]	**13.** [0.4]	**14.** [6.023×10^{24}]		**32.** (*a*)	**33.** (*a*)	**34.** (*a*) **35.** (*c*)
36. (*b, c*)	**37.** [222]	**38.** [2.385]	**39.** [6.47]	**40.** [495]	**41.** [3]	**42.** [7/3]	**43.** [4.14]	**44.** (C-12 isotopes)

2 ATOMIC STRUCTURE

JEE-Advanced

Subatomic Particles and Atomic Models

Single Correct

1. Rutherford's experiment, which established the nuclear model of the atom, used a beam of **(IIT JEE 2002)**
   (*a*) β-particles, which impinged on a metal foil and got absorbed
   (*b*) γ-rays, which impinged on a metal foil and got scattered
   (*c*) helium atoms, which impinged on a metal foil and got scattered
   (*d*) helium nuclei, which impinged on a metal foil and got scattered

2. Rutherford's alpha particle scattering experiment eventually led to the conclusion that **(IIT JEE 1986)**
   (*a*) mass and energy are related
   (*b*) electrons occupy space around the nucleus
   (*c*) neutrons are buried deep in the nucleus
   (*d*) the point of impact with matter can be precisely determined

3. The radius of an atomic nucleus is of the order of **(IIT JEE 1985)**
   (*a*) 10^{-10} cm (*b*) 10^{-13} cm
   (*c*) 10^{-15} cm (*d*) 10^{-18} cm

4. The increasing order (lowest first) for the values of e/m (charge/mass) for electron (e), proton (p), neutron (n) and alpha particle (α) is **(IIT JEE 1984)**
   (*a*) e, p, n, α (*b*) n, p, e, α
   (*c*) n, p, α, e (*d*) n, α, p, e

5. Rutherford's scattering experiment is related to the size of the **(IIT JEE 1983)**
   (*a*) nucleus (*b*) atom (*c*) electron (*d*) neutron

6. Rutherford's experiment on scattering of α-particles showed for the first time that the atom has **(IIT JEE 1981)**
   (*a*) electrons (*b*) protons (*c*) nucleus (*d*) neutrons

Multiple Correct

7. The atomic nucleus contains **(IIT JEE 1988)**
   (*a*) protons (*b*) neutrons
   (*c*) electrons (*d*) photons

8. The sum of the number of neutrons and proton in the isotopes of hydrogen is **(IIT JEE 1986)**
   (*a*) 6 (*b*) 5 (*c*) 4 (*d*) 3

9. When alpha particles are sent through a thin metal foil, most of them go straight through the foil, because **(IIT JEE 1984)**
   (*a*) alpha particles are much heavier than electrons
   (*b*) alpha particles are positively charged
   (*c*) most part of the atom is empty space
   (*d*) alpha particles move with high velocity

10. Many elements have non-integral atomic masses, because **(IIT JEE 1984)**
   (*a*) they have isotopes
   (*b*) their isotopes have non-integral masses
   (*c*) their isotopes have different masses
   (*d*) the constituents, neutrons, protons and electrons, combine to give fractional masses

11. An isotone of $^{76}_{32}Ge$ **(IIT JEE 1984)**
   (*a*) $^{77}_{32}Ge$ (*b*) $^{77}_{33}As$ (*c*) $^{77}_{34}Se$ (*d*) $^{78}_{34}Se$

Fill in the Blanks

12. Elements of the same mass number but of different atomic numbers are known as ..... **(IIT JEE 1983)**

13. The mass of a hydrogen is ..... kg. **(IIT JEE 1982)**

14. Isotopes of an element differ in the number of ..... in their nuclei. **(IIT JEE 1982)**

True/False

15. In a given electric field, β-particles are deflected more than α-particles in spite of α-particles having larger charge. **(IIT JEE 1993)**

Developments Leading to the Bohr's Model of Atom

Single Correct

16. The first use of quantum theory to explain the structure of atom was made by **(IIT JEE 1997)**
   (*a*) Heisenberg (*b*) Bohr
   (*c*) Planck (*d*) Einstein

17. Which of the following relates to photons both as wave motion and as a stream of particles? **(IIT JEE 1992)**
   (*a*) Interference (*b*) $E = mc^2$
   (*c*) Diffraction (*d*) $E = h\nu$

18. Which of the following does not characterize X-rays? **(IIT JEE 1992)**

(*a*) The radiation can ionise gases
(*b*) It causes *ZnS* to fluoresce
(*c*) Deflected by electric and magnetic fields
(*d*) Have wavelengths shorter than ultraviolet rays

19. The ratio of the energy of a photon of 2000Å wavelength radiation to that of 4000Å radiation is **(IIT JEE 1986)**

(*a*) $\frac{1}{4}$ (*b*) 4 (*c*) $\frac{1}{2}$ (*d*) 2

20. Electromagnetic radiation with maximum wavelength is **(IIT JEE 1985)**

(*a*) ultraviolet (*b*) radio wave
(*c*) X-ray (*d*) infrared

21. Bohr's model can explain **(IIT JEE 1985)**

(*a*) the spectrum of hydrogen atom only
(*b*) spectrum of an atom or ion containing one electron only
(*c*) the spectrum of hydrogen molecule
(*d*) the solar spectrum

Numerical/Integer Types

22. The atomic masses of He and Ne are 4 and 20 amu, respectively. The value of the de-Broglie wavelength of He gas at – 73°C is 'M' times that of the de-Broglie wavelength of Ne at 727°C. M is

C-51.1 W-21.28 UA-27.62 (JEE Adv. 2013)

23. The work function (ϕ) of some metals is listed below. The number of metals which will show photoelectric effect when light of 300 nm wavelength falls on the metal is **(IIT JEE 2011)**

Metal	Li	Na	K	Mg	Cu	Ag	Fe	Pt	W
Φ(eV)	2.4	2.3	2.2	3.7	4.8	4.3	4.7	6.3	4.75

Fill in the Blanks

24. The light radiations with discrete quantities of energy are called ..... **(IIT JEE 1993)**

True/False

25. Gamma rays are electromagnetic radiations of wavelengths of 10^{-6} to 10^{-5} cm **(IIT JEE 1983)**

Subjective

26. Calculate the wave number for the shortest wavelength transition in the Balmer series of atomic hydrogen. **(IIT JEE 1996)**

27. Iodine molecule dissociates into atoms after absorbing light to 4500Å. If one quantum of radiation is absorbed by each molecule, calculate the kinetic energy of iodine atoms. (Bond energy of I_2 = 240 kJ mol^{-1}) **(IIT JEE 1995)**

28. What transition in the hydrogen spectrum would have the same wavelength as the Balmer transition n = 4 to n = 2 of He^+ spectrum? **(IIT JEE 1993)**

Bohr's Model for Hydrogen Atom

Single Correct

29. The kinetic energy of an electron in the second Bohr orbit of a hydrogen atom is (a_0 is Bohr radius).

C-15.62 W-49.46 UA - 34.92 (IIT JEE 2012)

(*a*) $\frac{h^2}{4\pi^2 m a_0^2}$ (*b*) $\frac{h^2}{16\pi^2 m a_0^2}$
(*c*) $\frac{h^2}{32\pi^2 m a_0^2}$ (*d*) $\frac{h^2}{64\pi^2 m a_0^2}$

30. Which hydrogen like species will have same radius as that of Bohr orbit of hydrogen atom? **(IIT JEE 2004)**

(*a*) n = 2, Li^{2+} (*b*) n = 2, Be^{3+}
(*c*) n = 2, He^+ (*d*) n = 3, Li^{2+}

31. The energy of an electron in the first Bohr orbit of H-atom is – 13.6eV. The possible energy value(s) of the excited state(s) for electrons in Bohr orbits of hydrogen is (are) **(IIT JEE 1998)**

(*a*) –3.4eV (*b*) –4.2eV
(*c*) –6.8eV (*d*) +6.8eV

32. The wavelength of a spectral line for an electronic transition is inversely related to **(IIT JEE 1988)**

(*a*) the number of electrons undergoing the transition
(*b*) the nuclear charge of the atom
(*c*) the difference in the energy of the energy levels involved in the transition
(*d*) the velocity of the electron undergoing the transition

33. Which electronic level would allow the hydrogen atom to absorb a photon but not to emit a photon? **(IIT JEE 1984)**

(*a*) 3s (*b*) 2p
(*c*) 2s (*d*) 1s

Comprehensive/Passage Based

Consider the Bohr's model of a one-electron atom where the electron moves around the nucleus. In the following List-I contains some quantities for the nth orbit of the atom and List-II contains options showing how they depend on n.

	List-I		List-II
(I)	Radius of the nth orbit	(P)	$\propto n^{-2}$
(II)	Angular momentum of the electron in the nth orbit	(Q)	$\propto n^{-1}$
(III)	Kinetic energy of the electron in the nth orbit	(R)	$\propto n^{0}$
(IV)	Potential energy of the electron in the nth orbit	(S)	$\propto n^{1}$
		(T)	$\propto n^{2}$
		(U)	$\propto n^{1/2}$

34. Which of the following options has the correct combination considering List-I and List-II?

C-75.83 W-17.64 UA-6.53 (JEE Adv. 2019)

(*a*) (II), (R) (*b*) (I), (P)
(*c*) (I), (T) (*d*) (II), (Q)

35. Which of the following options has the correct combination considering List-I and List-II?

C-70.33 W-20.49 UA-9.19 (JEE Adv. 2019)

(*a*) (III), (S) (*b*) (IV), (Q)
(*c*) (IV), (U) (*d*) (III), (P)

36. According to Bohr's theory,

E_n = Total energy K_n = Kinetic Energy V_n = Potential Energy r^n = Radius of nth orbit **(IIT JEE 2006)**

	Column-I		Column-II
A.	$V_n / K_n = ?$	p	0
B.	If radius of *n*-th orbit $\propto E_n^x, x = ?$	q	–1
C.	Angular momentum in lowest	r.	–2
D.	$\frac{1}{r^n} \propto Z^y, y = ?$	s	1

Numerical/Integer Types

37. For He^+, a transition takes place from the orbit of radius 105.8 pm to the orbit of radius 26.45 pm. The wavelength (in nm) of the emitted photon during the transition is ___.

[Use: Bohr radius, a = 52.9 pm

Rydberg constant, $R_H = 2.2 \times 10^{-18}$ J

Planck's constant, $h = 6.6 \times 10^{-34}$ J s

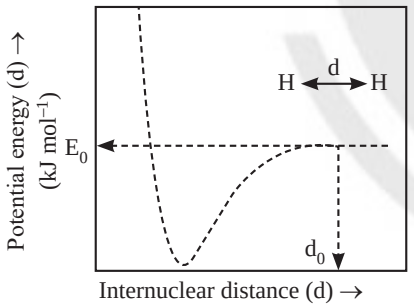
Speed of light, $c = 3 \times 10^8$ m s^{-1}]

C-11.93 W-66.52 UA-21.55 (JEE Adv. 2023)

38. The figure below is the plot of potential energy versus internuclear distance (d) of H_2 molecule in the electronic ground state. What is the value of the net potential energy E_0 (as indicated in the figure) in kJ mol^{-1}, for d = d_0 at which the electron-electron repulsion and the nucleus-nucleus repulsion energies are absent? As reference, the potential energy of H atom is taken as zero when its electron and the nucleus are infinitely far apart.

Use Avogadro constant as 6.023×10^{23} mol^{-1}.

C-0.14 W-65.68 UA-34.18 (JEE Adv. 2020)



Fill in the Blanks

39. 8 *g* each of oxygen and hydrogen at 27°C will have the total kinetic energy in the ratio of ..... **(IIT JEE 1989)**

Subjective

40. (*a*) Calculate velocity of electron in first Bohr orbit of hydrogen atom (Given r = a_0).

(*b*) Find de-Broglie wavelength of the electron in first Bohr orbit.

(*c*) Find the orbital angular momentum of 2*p*-orbital in terms of $h / 2\pi$ units. **(IIT JEE 2005)**

41. The wavelength corresponding to maximum energy for hydrogen is 91.2 nm. Find the corresponding wavelength for He^+ ion **(IIT JEE 2003)**

42. Calculate the energy required to excite 1 L of hydrogen gas at 1 atm and 298 K to the first excited state of atomic hydrogen. The energy for the dissociation of H – H bond is 436 kJ mol^{-1}. **(IIT JEE. 2000)**

43. With what velocity should an α-particle travel towards the nucleus of a copper atom so as to arrive at a distance 10^{-13} *m* from the nucleus of the copper atom? **(IIT JEE 1997)**

44. Consider the hydrogen atom to be proton embedded in a cavity of radius a_0 (Bohr's radius) whose charge is neutralised by the addition of an electron to the cavity in vacuum, infinitely slowly. Estimate the average total energy of an electron in its ground state in a hydrogen atom as the work done in the above neutralisation process. Also, if the magnitude of the average kinetic energy is half the magnitude of the average potential energy, find the average potential energy. **(IIT JEE 1996)**

45. Find out the number of waves made by a Bohr's electron in one complete revolution in its 3rd orbit. **(IIT JEE 1994)**

46. Estimate the difference in energy between 1st and 2nd Bohr's orbit for a hydrogen atom. At what minimum atomic number, a transition from *n* = 2 to *n* = 1 energy level would result in the emission of X-rays with $l = 3.0 \times 10^{-8}$ *m*? Which hydrogen atom-like species does this atomic number correspond to? **(IIT JEE 1993)**

47. According to Bohr's theory, the electronic energy of hydrogen atom in the nth Bohr's orbit is given by:

$$E_n = \frac{-21.7 \times 10^{-19}}{n^2} J$$

Calculate the longest wavelength of electron from the third Bohr's orbit of the He^+ ion **(IIT JEE 1990)**

48. The electron energy in hydrogen atom is given by $E_n = -\frac{21.7 \times 10^{-12}}{n^2}$ erg. Calculate the energy required to remove an electron completely from the n = 2 orbit. What is the longest wavelength (in cm) of light that can be used to cause this transition? **(IIT JEE 1984)**

49. Calculate the wavelength in Angstroms of the photon that is emitted when an electron in the Bohr's orbit, *n* = 2 returns to the orbit, *n* = 1 in the hydrogen atom. The ionisation potential of the ground state hydrogen atom is 2.17×10^{-11} *erg* per atom. **(IIT JEE 1982)**

50. The energy of the electron in the second and third Bohr's orbits of the hydrogen atom is -5.42×10^{-12} erg and -2.41×10^{-12} erg respectively. Calculate the wavelength of the emitted light when the electron drops from the third to the second orbit. **(IIT JEE 1981)**

Towards Quantum Mechanical Model of the Atom

Single Correct

51. The wavelength associated with a golf ball weighing 200 g and moving at a speed of 5 m / h is of the order **(IIT JEE 2001)**

(*a*) 10^{-10} m (*b*) 10^{-20} m

(*c*) 10^{-30} m (*d*) 10^{-40} m

Numerical/Integer Types

52. Consider a Helium (He) atom that absorbs a photon of wavelength 330 nm. The change in the velocity (in cm s^{-1}) of He atom after the photon absorption is _____.

(Assume: Momentum is conserved when photon is absorbed).

Use: Planck constant = 6.6×10^{-34} Js, Avogadro number = 6×10^{23} mol^{-1}, molar mass of He = 4 g mol^{-1}

C-7.56 W-73.79 UA-18.65 (JEE Adv. 2021)

Fill in the Blanks

53. The uncertainty principle and the concept of wave nature of matter were proposed by ..... and ..... respectively **(IIT JEE 1988)**

Subjective

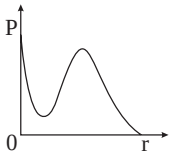
54. An electron beam can undergo diffraction by crystals. Through what potential should a beam of electrons be accelerated so that its wavelength becomes equal to 1.54Å. **(IIT JEE 1997)**

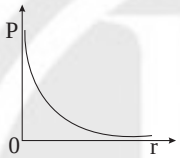
Quantum Mechanical Model of the Atom

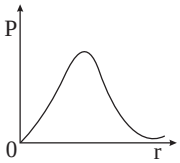
Single Correct

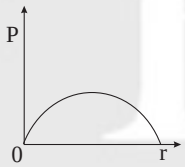
55. P is the probability of finding the 1s electron of hydrogen atom in a spherical shell of infinitesimal thickness, dr, at a distance r from the nucleus. The volume of this shell is $4\pi r^2$ dr. The qualitative sketch of the dependence of P on r is

C-37.22 W-37.41 UA-25.38 (JEE Adv. 2016)

(*a*) 

(*b*) 

(*c*) 

(*d*) 

56. The number of radial nodes in 3s and 2p respectively are **(IIT JEE 2005)**

(*a*) 2 and 0 (*b*) 0 and 2 (*c*) 1 and 2 (*d*) 2 and 1

57. If the nitrogen atom had electronic configuration $1s^7$, it would have energy lower than that of the normal ground state configuration $1s^2 2s^2 2p^3$, because the electrons would be closer to the nucleus, yet $1s^7$ is not observed, because it violates **(IIT JEE 2002)**

(*a*) Heisenberg uncertainty principle
(*b*) Hund's rule
(*c*) Pauli exclusion principle
(*d*) Bohr postulate of stationary orbits

58. The quantum numbers $+\frac{1}{2}$ and $-\frac{1}{2}$ for the electron spin represent **(IIT JEE 2001)**

(*a*) rotation of the electron in clockwise and anti-clockwise direction respectively
(*b*) rotation of the electron in anti-clockwise and clockwise direction respectively
(*c*) magnetic moment of the electron pointing up and down respectively
(*d*) two quantum mechanical spin states which have no classical analogue

59. The number of nodal planes in a p_x orbital is **(IIT JEE 2001)**

(*a*) one (*b*) two (*c*) three (*d*) zero

60. The electronic configuration of an element is $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^5, 4s^1$. This represents its **(IIT JEE 2000)**

(*a*) excited state (*b*) ground state
(*c*) cationic form (*d*) anionic form

61. The electrons, identified by quantum numbers n and *l*, (i) $n = 4$, $l = 1$, (ii) n = 4, $l = 0$, (iii) n = 3, $l = 2$, (iv) n = 3, $l = 1$ can be placed in order of increasing energy, from the lowest to highest, as **(IIT JEE 1999)**

(*a*) (iv) < (ii) < (iii) < (i) (*b*) (ii) < (iv) < (i) < (iii)
(*c*) (i) < (iii) < (ii) < (iv) (*d*) (iii) < (i) < (iv) < (ii)

62. For a d-electron, the orbital angular momentum is **(IIT JEE 1997)**

(*a*) $\sqrt{6}\left(\frac{h}{2\pi}\right)$ (*b*) $\sqrt{2}\left(\frac{h}{2\pi}\right)$
(*c*) $\left(\frac{h}{2\pi}\right)$ (*d*) $2\left(\frac{h}{2\pi}\right)$

63. The orbital angular momentum of an electron in 2*s*-orbital is **(IIT JEE 1996)**

(*a*) $+\frac{1}{2}\cdot\frac{h}{2\pi}$ (*b*) zero
(*c*) $\frac{h}{2\pi}$ (*d*) $\sqrt{2}\cdot\frac{h}{2\pi}$

64. The outermost electronic configuration of the most electronegative element is **(IIT JEE 1990)**

(*a*) $ns^2 np^3$ (*b*) $ns^2 np^4$
(*c*) $ns^2 np^5$ (*d*) $ns^2 np^6$

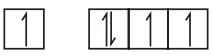
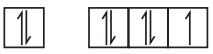
65. The correct set of quantum numbers for the unpaired electron of chlorine atom is **(IIT JEE 1989)**

(*a*) n = 2, $\ell = 1$, m = 0 (*b*) n = 2, $\ell = 1$, m = 1
(*c*) n = 3, $\ell = 1$, m = 1 (*d*) n = 3, $\ell = 0$, m = 0

66. The correct ground state electronic configuration of chromium atom is **(IIT JEE 1989)**

(*a*) $[Ar]3d^5 4s^1$ (*b*) $[Ar]3d^4 4s^2$
(*c*) $[Ar]3d^6 4s^0$ (*d*) $[Ar]4d^5 4s^1$

67. The orbital diagram in which the Aufbau principle is violated **(IIT JEE 1988)**

(*a*) (*b*) 
(*c*) (*d*) 

68. Which one of the following sets of quantum numbers represents an impossible arrangement for n, l, m, s? **(IIT JEE 1986)**

(*a*) $\left(3, 2, -2, \frac{1}{2}\right)$ (*b*) $4, 0, 0, \frac{1}{2}$
(*c*) $3, 2, -3, \frac{1}{2}$ (*d*) $5, 3, 0, -\frac{1}{2}$

69. Correct set of four quantum numbers for the valence (outermost) electron of rubidium ($Z = 37$) is **(IIT JEE 1984)**

(*a*) $5, 0, 0, +\frac{1}{2}$ (*b*) $5, 0, 1, +\frac{1}{2}$
(*c*) $5, 1, 1, +\frac{1}{2}$ (*d*) $6, 0, 0, +\frac{1}{2}$

70. The principal quantum number of an atom is related to the **(IIT JEE 1983)**

(*a*) size of the orbital
(*b*) spin angular momentum
(*c*) orientation of the orbital in space
(*d*) orbital angular momentum

71. Any *p*-orbital can accommodate upto **(IIT JEE 1983)**

(*a*) four electrons
(*b*) six electrons
(*c*) two electrons with parallel spins
(*d*) two electrons with opposite spins

Multiple Correct

72. The ground state energy of hydrogen atom is –13.6 eV. Consider an electronic state of He^+ whose energy, azimuthal quantum number and magnetic quantum number are –3.4 eV, 2 and 0, respectively.

Which of the following statement(s) is (are) true for the state Ψ?

C-23.82 W-41.54 UA-20.07 PC-14.57 (JEE Adv. 2019)

(*a*) It is a 4 d state

(*b*) The nuclear charge experienced by the electron in this state is less than 2e, where e is the magnitude of the electronic charge

(*c*) It has 2 angular nodes

(*d*) It has 3 radial nodes

73. The ground state electronic configuration of nitrogen atom can be represented by **(IIT JEE 1999)**

(*a*) ⇅ ⇅ ↑ ↑ ↑ (*b*) ⇅ ⇅ ↑ ↓ ↑

(*c*) ⇅ ⇅ ↑ ↓ ↓ (*d*) ⇅ ⇅ ↓ ↓ ↓

74. Which of the following statement (s) is (are) correct? **(IIT JEE 1998)**

(*a*) The electronic configuration of Cr is $[Ar]3d^5 4s^1$ (atomic number of Cr = 24)

(*b*) The magnetic quantum number may have a negative value

(*c*) In silver atom, 23 electrons have a spin of one type and 24 of the opposite type. (Atomic number of Ag = 47)

(*d*) The oxidation state of nitrogen in HN_3 is –3

Assertion Reason/Statement Based

Read the following questions and answer as per the direction given below:

(*a*) Statement-I is correct; Statement-II is correct Statement-II is the correct explanation of Statement-I

(*b*) Statement-I is correct; Statement-II is correct Statement-II is not the correct explanation of Statement-I

(*c*) Statement-I is correct; Statement-II is incorrect

(*d*) Statement-I is incorrect; Statement-II is correct

75. **Statement-I:** The first ionisation energy of Be in greater than that of B.

Statement-II: 2p-orbital is lower in energy than 2s. **(IIT JEE 2000)**

Comprehension/Passage Based

The hydrogen-like species Li^{2+} is in a spherically symmetric state S_1 with one radial node. Upon absorbing light, the ion undergoes transition to a state S_2. The state S_2 has one radial node and its energy is equal to the ground state energy of the hydrogen atom.

76. The state S_1 is **(IIT JEE 2010)**

(*a*) 1s (*b*) 2s (*c*) 2p (*d*) 3s

77. Energy of the state S_1 in units of the hydrogen atom ground state energy is **(IIT JEE 2010)**

(*a*) 0.75 (*b*) 1.50 (*c*) 2.25 (*d*) 4.50

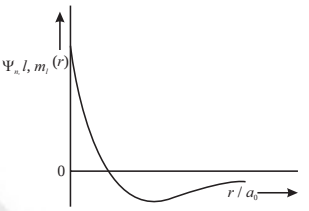
78. The orbital angular momentum quantum number of the state S_2 is **(IIT JEE 2010)**

(*a*) 0 (*b*) 1 (*c*) 2 (*d*) 3

Match the Column

The wave function, ψ_n, l, m_l is a mathematical function whose value depends upon spherical polar coordinates (r, θ, ϕ) of the electron and characterised by the quantum number n, l and m_l. Here r is distance from nucleus, θ is colatitude and ϕ is azimuth. In the mathematical functions given in the Table, Z is atomic number and a_0 is Bohr radius.

	Column-I		Column-II
(I)	Is-orbital	(i)	$\psi_{n,l,m_l} \propto \left(\frac{Z}{a_0}\right)^{\frac{3}{2}} e^{-\left(\frac{Zr}{a_0}\right)}$
(II)	2s-orbital	(ii)	one radial node
(III)	$2p_z$ - orbital	(iii)	$\grave{E}_{n,l,m_l} \propto \left(\frac{Z}{a_0}\right)^{\frac{5}{2}} re^{-\left(\frac{Zr}{a_0}\right)} \cos$,
(IV)	$3d_z^2$ – orbital	(iv)	xy-plane is a nodal plane

	Column-III
(P)	
(Q)	Probability density at nucleus $\propto \frac{1}{a_0^3}$
(R)	Probability density is maximum at nucleus
(S)	Energy needed to excite electron from n = 2 state to n = 4 state is $\frac{27}{32}$ times the energy needed to excite electron from *n* = 2 state to n = 6.

79. For He^+ ion, the only INCORRECT combination is

C-16.69 W-40.53 UA-42.78 (JEE Adv. 2017)

(*a*) (I) (i) (S) (*b*) (II) (ii) (Q)

(*c*) (I) (iii) (R) (*d*) (I) (i) (R)

80. For the given orbital in Column-I, the only CORRECT combination for nay hydrogen-like species is

C-28.59 W-30.39 UA-41.01 (JEE Adv. 2017)

(*a*) (II) (ii) (P) (*b*) (I) (ii) (S)

(*c*) (IV) (iv) (R) (*d*) (III) (iii) (P)

81. For hydrogen atom, the only CORRECT combination is

C-32.01 W-30.18 UA-37.8 (JEE Adv. 2017)

(*a*) (I) (i) (P) (*b*) (I) (iv) (R)

(*c*) (II) (i) (Q) (*d*) (I) (i) (S)

Numerical/Integer Types

82. Not considering the electronic spin, the degeneracy of the second excited state (*n* = 3) of H-atom is 9 , while the degeneracy of the second excited state of H^- is **(JEE Adv. 2015)**

83. In an atom, the total number of electrons having quantum numbers $n = 4, |m_l| = 1$ and $m_s = -\frac{1}{2}$ is

C-26.82 W-69.72 UA- 3.46 (JEE Adv. 2014)

84. The maximum number of electrons that can have principal quantum number, *n* = 3 and spin quantum number, m_s = –1/2, is

(IIT JEE 2011)

Fill in the Blanks

85. The outermost electronic configuration of Cr is _____. **(IIT JEE 1994)**

86. Wave functions of electrons in atoms and molecules are called _____. **(IIT JEE 1993)**

87. The $2p_x$, $2p_y$ and $2p_z$ orbitals of atom have identical shapes but differ in their _____. **(IIT JEE 1993)**

88. When there are two electrons in the same orbital, they have _____ spins **(IIT JEE 1983)**

True/False

89. The electron density in the *XY*-plane in $3d_{x^2-y^2}$ orbital zero. **(IIT JEE 1986)**

90. The energy of the electron in the 3d-orbital is less than that in the 4*s* orbital in the hydrogen atom. **(IIT JEE 1983)**

91. The outer electronic configuration of the ground state chromium atom is $3d^4 4s^2$. **(IIT JEE 1982)**

Subjective

92. The Schrodinger wave equation for hydrogen atom is

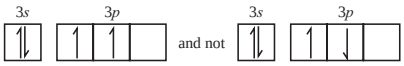
$$\psi_{2s} = \frac{1}{4(2\pi)^{1/2}}\left(\frac{1}{a_0}\right)^{3/2}\left(2-\frac{r}{a_0}\right)e^{-r/2a_0}$$

where, a_0 is Bohr's radius. Let the radial node in 2s be at r_0. Then, find r in terms of a_0.

(*b*) A base ball having mass 100 g moves with velocity 100 m/s. Find out the value of wavelength of baseball. **(IIT JEE 2004)**

93. What is the maximum number of electrons that may be present in all the atomic orbitals with principal quantum number 3 and azimuthal quantum number 2? **(IIT JEE 1985)**

94. Give reason why the ground state outermost electronic configuration of silicon is



(IIT JEE 1985)

ANSWER KEY

1. (*d*)	**2.** (*b*)	**3.** (*b*)	**4.** (*d*)	**5.** (*a*)	**6.** (*c*)	**7.** (*a, b*)	**8.** (*a*)	**9.** (*a, c*)	**10.** (*a,c*)
11. (*b, d*)	**12.** Isobar	**13.** 1.66×10^{-24} kg		**14.** Neutrons	**15.** TRUE	**16.** (*b*)	**17.** (*d*)	**18.** (*c*)	**19.** (*d*)
20. (*b*)	**21.** (*b*)	**22.** [5]	**23.** [4]	**24.** Photons	**25.** FALSE	**29.** (*c*)	**30.** (*b*)	**31.** (*a*)	**32.** (*c*)
33. (*d*)	**34.** (*c*)	**35.** (*d*)	**36.** A→(r); B→(q); C→(p); D→(s)			**37.** [30]	**38.** [–5242.4192]		**39.** [1:16]
51. (*c*)	**52.** [30]	**53.** Heisenberg and De-Broglie			**55.** (*c*)	**56.** (*a*)	**57.** (*c*)	**58.** (*d*)	**59.** (*a*)
60. (*b*)	**61.** (*a*)	**62.** (*a*)	**63.** (*b*)	**64.** (*c*)	**65.** (*c*)	**66.** (*a*)	**67.** (*b*)	**68.** (*c*)	**69.** (*a*)
70. (*a*)	**71.** (*d*)	**72.** (*a*, c)	**73.** (*a, d*)	**74.** (*a, b, c*)	**75.** (*c*)	**76.** (*b*)	**77.** (*c*)	**78.** (*b*)	**79.** (*c*)
80. (*a*)	**81.** (*d*)	**82.** [3]	**83.** [6]	**84.** [9]	**85.** $[Ar]3d^5 4s^1$		**86.** Orbital	**87.** Orientation in space	
88. opposite	**89.** FALSE	**90.** TRUE	**91.** FALSE						

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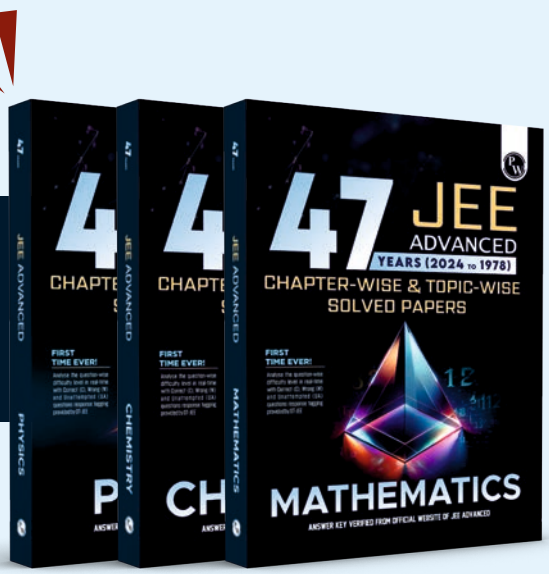

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